

Cognitive Skills and the Timing of First Births*

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Abstract

Among U.S. women born 1957–64, 63% of the lowest cognitive-skill quartile give birth before 22, against 18% of the highest. The gap is in unintended births, opens in high school, survives family background. In a life-cycle model where skills shift adherence—reliability of use—beyond wages and schooling, common adherence is rejected: it fits neither first-birth timing by skill nor within-method failure. Equalizing adherence cuts births by 21 by 16%, mostly for low-skill women; removing contraception’s cost does nearly as much, but its gains go to the high-skilled. A long-acting method at Colorado’s take-up has an effect the program’s size.

JEL codes: J13, J12, I21, J24, C61.

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1 Introduction

Early childbearing varies sharply with cognitive ability. In the National Longitudinal Survey of Youth 1979 (NLSY79), whose respondents were born between 1957 and 1964, 63 percent of women in the lowest quartile of the Armed Forces Qualification Test (AFQT) have had a first birth before age 22, against 18 percent in the highest. The gap is one of *timing*: first-birth hazards converge by the late twenties, so mean age at first birth differs by 5.5 years across quartiles while the share who ever become mothers differs far less.

This paper asks whether the standard opportunity-cost channel can account for the gap, and finds that it cannot. I first show that the gap is not an artifact of education sorting or of contraceptive method selection. I then estimate a dynamic life-cycle model of schooling, work, marriage and contraceptive method choice in which cognitive ability can shift *adherence*—how reliably a chosen method is used. The data reject the version of the model in which adherence is common across ability groups, even when wages, schooling, marriage, education-specific fertility control and ability-specific preferences for motherhood are all free to adjust. That version cannot reproduce the timing of first births by ability: with one adherence level per age it spreads teen births up the ability distribution, raising the highest quartile’s share with a first birth by 17 to 4.6 percent against 3.4 in the data and 3.3 in the estimated model, and the timing of first births alone accounts for 81 percent of the rejection; it cannot produce the ability gradient in failure rates among users of the same method; it recovers only 79 percent of the gradient in unintended first births by 21; and it distorts labor supply to compensate. Matching the data requires an ability shifter in the mapping from a chosen method to realized births—a fertility-control channel absent from existing structural models of fertility—and the welfare gains from acting on it are concentrated among the least advantaged women.

The question matters because early childbearing reduces women’s education, labor supply and lifetime earnings (Adda et al., 2017), delayed motherhood raises children’s human capital (Miller, 2009), and financially constrained young parents under-invest in their children (Caucutt and Lochner, 2020). Because early births concentrate among disadvantaged women, ability-dependent fertility timing is also a channel through which inequality is trans-

mitted across generations (Aizer et al., 2022; Seshadri and Zhou, 2022). About 45 percent of U.S. pregnancies were unintended as of 2011 (Finer and Zolna, 2016), and birth rates among women without a high-school degree have fallen markedly over recent decades (Kearney et al., 2022)—which makes how tightly fertility control binds for them a question with consequences well beyond the women themselves. Whether policy should act on the incentive to delay or on the means depends on which of the two binds.

The reduced-form evidence points to fertility control. The gap consists of births women report not having wanted at the time: by age 41, women in the lowest and highest ability quartiles are about equally likely to have had an intended first birth, and the whole difference is in unintended ones. Of the three margins of birth risk, ability predicts the two a woman controls—whether she uses a method and how well it works—and not the third, fecundity and exposure, measured by the birth rate of women using no method. Women in the lowest quartile are 14.7 percentage points less likely than those in the highest to be using any contraceptive method, and among women using the same method their annual birth rate is about 60 percent higher than in the highest quartile (3.6 against 2.2 percent, holding education, parity and age fixed). Among women using no method, the birth rate shows no ability gradient.

The gap is not education sorting: it opens at ages 14–17, when 92 percent of childless girls are still in high school, and it survives holding their grade and expected schooling fixed. And it is not the choice of method: low-ability women who contracept rely more on the pill, not less, and holding the method fixed widens the gap, from 1.2 to 1.4 points in the same sample.

The paper models live births and abstracts from abortion. Differential abortion cannot *generate* the gradient: doing so would require top-quartile women to terminate 2.04 times as many pregnancies as bottom-quartile women by age 25, against an observed ratio of 1.18 (see also Woodley of Menie et al., 2017). Because abortion is if anything more common at the top, the live-birth gap is somewhat wider than the conception gap, by about a fifth.¹

To interpret these patterns, I build a life-cycle model in which cognitive ability is an early,

¹Reported abortions per woman by age 25 *rise* weakly across AFQT quartiles, from 0.160 to 0.189; the income gradient in abortion that Jones and Jerman (2017) document does not transfer to cognitive ability in this cohort.

persistent trait² and women jointly choose schooling, labor supply, experience, marriage and—while childless and at risk—a contraceptive method from a menu of known efficacy. The innovation is in what follows that choice: a method’s failure rate depends on adherence, the reliability with which it is used, which is what separates perfect-use from typical-use failure rates ([Trussell, 2011](#)), and cognitive ability shifts adherence in addition to its effects through schooling and wages. The estimation also disciplines which births were wanted: the value of a birth varies across otherwise identical women, and the estimation targets the share of first births that were intended by age. This is what gives the test its power: when adherence does not vary with ability, the model reduces to the opportunity-cost explanation, and the only ways left to it to hold down the top quartile’s early births—one adherence level for every quartile, a larger career cost of a child, and a failure rate that is flat across ability within a method—are the ones the timing, employment and within-method moments reject.

Three counterfactuals sharpen the policy reading. Equalizing adherence to high-ability levels lowers first births by age 21 by 16 percent and before 18 by 24 percent, and its welfare gains accrue to the three lowest quartiles and not at all to the highest. Removing the cost of using contraception for every woman, an upper bound on what any subsidy can do, lowers first births by 21 by 14 percent, about as much as closing the adherence gap, but its gains go to the top of the ability distribution: most of what it is worth is a transfer to women already using a method, and even net of that transfer the highest quartile gains more than the lowest and the lowest gains least of all. Adding a long-acting reversible contraceptive (LARC), which removes the need for daily adherence, to the menu at a cost that reproduces the take-up of Colorado’s program lowers first births by 21 by 7 percent, of the same order as the program’s estimated effect on teen birth rates ([Lindo and Packham, 2017](#)), with the largest gains in the lowest quartile and the smallest in the highest; offered at the pill’s cost, so that the two clinical methods differ only in the adherence margin, it is taken up by 54 percent of users and lowers first births by 21 by 13 percent, still short of the 16 percent from closing the adherence gap. The shortfall is take-up, not technology: the method removes the adherence margin only for the women who adopt it. Evaluations of LARC programs find

²Measured cognitive ability is highly stable from late adolescence through adulthood ([Almlund et al., 2011](#)).

the same incidence, concentrated among younger and poorer women ([Lindo and Packham, 2017](#); [Secura et al., 2014](#)). These estimates come from a cohort that formed its contraceptive behavior when long-acting methods were effectively off the menu,³ so they measure what was foregone rather than what remains to be gained, and the Colorado comparison checks only the order of magnitude. The mechanism carries over: wherever effectiveness depends on what the user does, differences in the capacity to control fertility produce differences in birth timing that the cost of contraception cannot undo.

The paper contributes to the structural literature that, building on [Becker \(1960\)](#) and [Willis \(1973\)](#), models birth timing as a life-cycle decision ([Adda et al., 2017](#); [Caucutt et al., 2002](#); [Francesconi, 2002](#); [Heckman and Walker, 1990](#); [Hotz and Miller, 1988](#); [Wolpin, 1984](#)), and in [Sheran \(2007\)](#) and [Keane and Wolpin \(2010\)](#) jointly with schooling, work and marriage.⁴ Imperfect fertility control is not new to this literature: [Hotz and Miller \(1988\)](#) estimate a contraceptive choice index, [Sheran \(2007\)](#) makes births stochastic conditional on contraceptive use, [Choi \(2017\)](#) adds conception risk and abortion, [Ejrnaes and Jørgensen \(2020\)](#) model family planning under uninsurable income risk, and [Amador \(2017\)](#) the policy environment for abortion and contraception. What they share is the mapping from fertility-control behavior to conception: it varies with age, education or marital status, but not with a woman’s ability. The household-production literature came closer: [Rosenzweig and Schultz \(1985\)](#) estimate heterogeneity across couples in fecundity and in the efficiency of contraception, and [Rosenzweig and Schultz \(1989\)](#) show that schooling raises that efficiency; letting cognitive ability shift that mapping, net of schooling and inside a life-cycle model of schooling, work, marriage and fertility, is what is new here, and within this family of models the nested test rejects its removal: the career-cost incentives of [Adda et al. \(2017\)](#) and education-specific contraception costs are both present in the estimated model, and neither reproduces the gradient without it.

The paper also joins two literatures that have run in parallel. One documents that cognitive and noncognitive skills predict early parenthood ([Fé et al., 2022](#); [Heckman et al.,](#)

³NLSY79 women are at risk between the early 1970s and the late 1980s, when IUD use had collapsed after the withdrawal of the Dalkon Shield and the modern implant did not yet exist.

⁴[Doepke et al. \(2023\)](#) survey the economics of fertility, and [Kearney and Levine \(2026\)](#) the recent decline in high-income countries.

2006; Shearer et al., 2002) but does not measure the mechanism: Fé et al. (2022) find that a standard deviation of cognitive ability at age eight lowers the probability of being a parent by age 25 by 2.3 percentage points against a base rate of 12 percent, and that education mediates only about a quarter of the comparable effect they do decompose. The other studies fertility control: how the price and availability of contraception shape births and human-capital investment (Ananat and Hungerman, 2012; Bailey, 2006, 2010; Bailey et al., 2012, 2023; Goldin and Katz, 2002; Guldi, 2008; Kearney and Levine, 2009; Kelly et al., 2020; Myers, 2017), what drives teen childbearing (Di Nola et al., 2025; Kearney and Levine, 2012, 2015), and, closest to this paper, that more schooled couples use more efficiently the methods that leave room for misuse (Rosenzweig and Schultz, 1989), that the education gradient in fertility runs through unintended births rather than desired family size or opportunity costs (Musick et al., 2009), and that typical-use failure rates within a method vary with income and other user characteristics (Kost et al., 2008; Sundaram et al., 2017). Health economics supplies the framing: schooling and ability raise the output obtained from a given health input (Grossman, 2006; Michael, 1973; Rosenzweig and Schultz, 1989), patient self-management explains part of the socioeconomic gradient in health, and technologies that depend on the user’s effort widen that gradient while technologies that do not compress it (Goldman and Lakdawalla, 2005; Goldman and Smith, 2002). This paper connects them at the point where they meet: within a method and within education, skills change how well contraception works, not only whether it is used.⁵ That turns a predictive correlation into a parameter of the fertility-control problem, and one that policy can act on.

2 Data and Motivating Evidence

This section describes the data and shows that the ability gap in early first births is a gap in unintended births, that it opens before schooling diverges, and that it persists within contraceptive method.

⁵Noncognitive skills are the obvious competing measure. Controlling for the Rotter and Rosenberg scales moves no reduced-form coefficient by more than 11 percent (Online Appendix OA.5.4).

2.1 Data Description

The NLSY79 follows a nationally representative cohort born between 1957 and 1964 who were ages 14–22 at the first interview in 1979. Cognitive ability is proxied by the AFQT, a composite of the Armed Services Vocational Aptitude Battery administered early in the panel (e.g., Heckman et al., 2006; Neal and Johnson, 1996). The analysis uses the 5,820 women with a valid AFQT score whose schooling histories the model can represent.⁶

The within-method analysis uses the seven waves in which the survey records both contraceptive method and pregnancy wantedness—1982, 1984, 1985, 1986, 1988, 1990 and 1992—when the cohort is aged 18–35. The outcome is a contraceptive failure: a birth in the window counts as one only if the woman reports that she had not wanted the pregnancy at the time, since a woman who stopped using her method in order to conceive has not had it fail.⁷ Births are measured in a fixed 12-month window from July of year t through June of $t + 1$, constructed from child birth dates; because interviews were fielded between late January and mid-August, the method reported at the wave- t interview precedes the outcome. Methods are grouped into three tiers by typical-use efficacy—oral contraceptives, barrier methods and behavioral methods—with dual users assigned to the most effective tier. LARC and sterilization are excluded.⁸

2.2 Wantedness and the Nature of the Ability Gap

Figure 1 shows what kind of births make up the ability gap. For every woman in the sample it records whether she had a first birth by ages 17, 21 and 41, split by the wantedness she reported for that birth on its own pregnancy record: *unintended* if it came too soon or was not wanted at all, *intended* otherwise, including births conceived after stopping contraception in order to become pregnant. Low-ability women are not more likely to ever have an intended

⁶Online Appendix OA.1.3 documents the estimation samples and the quartile cut-points. The battery was fielded in summer 1980, by which time some women had already had a first birth; Online Appendix OA.5.3 reports how many, by quartile, and shows that the gradient survives dropping them and inverse-probability weighting for attrition; Online Appendix OA.5.1 shows it survives adjusting the score for age at test.

⁷Online Appendix OA.1.7 gives the full classification rule and the birth-record and pregnancy-record items it draws on.

⁸IUD uptake was negligible in this cohort after the Dalkon Shield recall, and sterilization removes a woman from the risk pool irrespective of ability. Online Appendix OA.1 gives the full construction.

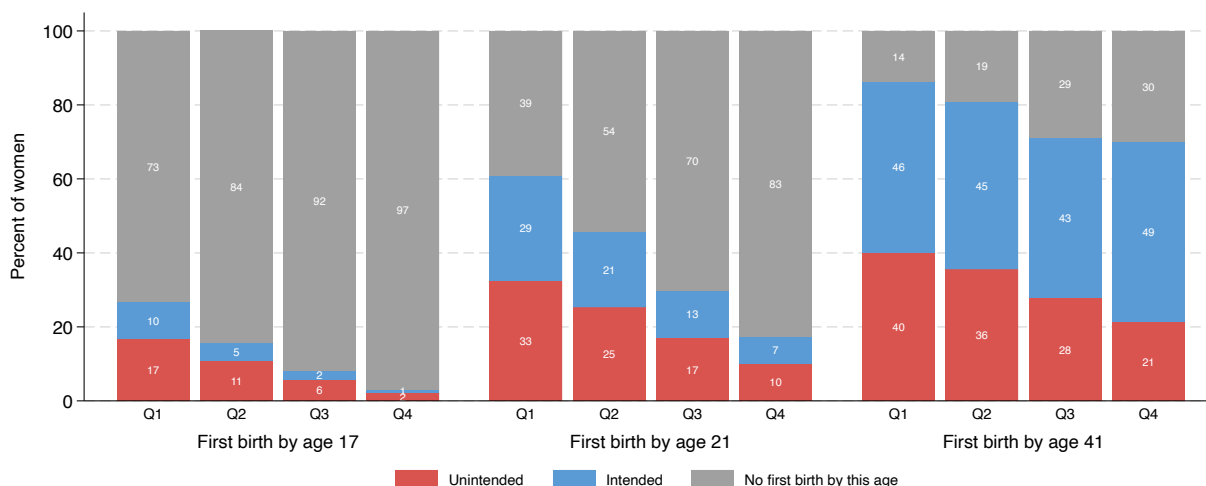


Figure 1. Intended and unintended first births by AFQT quartile. Share of women with a first birth by the age shown; each bar covers all women in the quartile, so the third segment is those who have not had a first birth by that age. A first birth is unintended if she reported wanting the pregnancy “not at that time” or “not at all”, or “didn’t matter” while using a method, and intended if she reported “yes”, “didn’t matter” while using none, or having stopped contraception in order to conceive. NLSY79; numbers and standard errors in Online Appendix Table OA.7.

first birth—by age 41, 46.2 percent of women in the lowest quartile and 48.7 percent in the highest have had one—but they are nearly twice as likely to have had an unintended one, 40.1 against 21.3 percent. The lifetime difference in first births between the bottom and top quartiles is therefore a difference in unintended births. It is also front-loaded: more than half of the gap in first births by age 21 has opened by 17, and 62 percent of the gap at 17 is unintended births; at 21 the share is 51 percent, because low-ability women also have their intended first births sooner.⁹ Musick et al. (2009) find the same for the education gradient in the NLSY79 and conjecture that it runs through how consistently contraception is used; I estimate that consistency as a parameter, within education and across ability.

⁹The figure reports the share of *women* who have had an unintended first birth, not the share of *births* that were unintended. The two differ because the number of early first births itself varies with ability: conditional on having a first birth at a given age, the unintended share does not differ by quartile (Online Appendix OA.3).

2.3 Ability and the Three Margins of Birth Risk

Two alternatives could produce the gradient without any difference in how well women use contraception: education sorting and contraceptive method selection. I take them in turn.

Is the ability gap an artifact of education sorting? AFQT predicts educational attainment, and education is associated with fertility through channels the model will interpret as opportunity costs.¹⁰ Comparing women of different ability with the same completed schooling does not work for the births that matter most: more than half of the gap in first births by age 21 has opened by 17 (Figure 1), while girls in every quartile are still in school, and a teen birth lowers the schooling a woman goes on to complete (Ashcraft et al., 2013; Fletcher and Wolfe, 2009; Klepinger et al., 1999), though Hotz et al. (2005) find much smaller effects once selection is addressed, so completed schooling is a “bad control” (Angrist and Pischke, 2009). Following the literature on first-birth hazards (Blossfeld and Huinink, 1991; Upchurch and McCarthy, 1990), I measure schooling as it stands before the conception, which the NLSY79’s annual interviews allow from age 14. For each childless woman aged 14–20 at an annual interview t , I estimate

$$F_{it} = \sum_{q=2}^4 \beta_q \mathbf{1}[\text{AFQT}_i = q] + \mathbf{s}'_{it} \boldsymbol{\gamma}_s + \mathbf{z}'_i \boldsymbol{\gamma}_z + \alpha_{\text{age}(i,t)} + \delta_t + \varepsilon_{it}, \quad (1)$$

where F_{it} equals one if her first child is born in calendar year $t + 1$, a birth conceived, almost always, after the interview; $\mathbf{s}_{it}$ is her schooling at the interview, enrollment status and highest grade completed, which precedes the conception; $\mathbf{z}_i$ holds what she reported in 1979, before any of these births: the schooling and the number of children she expected, her Rotter locus-of-control score and her parents’ schooling; $\alpha_{\text{age}(i,t)}$ and δ_t are age and year fixed effects, and standard errors are clustered by woman.

Table 1 reports the estimates. At ages 14–17, 92 percent of the woman-years are girls enrolled in high school: schooling has not yet diverged, and the gradient is already there. Among them, the top-quartile coefficient on the annual probability of a first birth is -9.4

¹⁰The quasi-experimental evidence on the causal effect of schooling on fertility is mixed (Fort et al., 2016; McCrary and Royer, 2011). Nothing below turns on a causal reading; the exercise asks whether the ability gradient survives when schooling is held fixed.

Table 1. The Ability Gradient in First Births Before Schooling Is Realized (LPM, 1979–1994)

	Ages 14–17					Ages 18–20	
	All		Enrolled in high school			All	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Any first birth: Q4 minus Q1	-9.5 (1.0)	-9.3 (1.1)	-9.4 (1.1)	-6.1 (1.2)	-5.0 (1.2)	-10.3 (1.0)	-7.0 (1.1)
Q1 mean	11.4	11.4	10.4	10.4	10.4	14.1	14.1
Unintended first birth: Q4 minus Q1	-6.1 (0.8)	-6.6 (0.9)	-6.8 (0.9)	-5.3 (1.0)	-3.6 (1.0)	-4.0 (0.7)	-2.9 (0.8)
Q1 mean	7.3	7.3	7.1	7.1	7.1	6.1	6.1
Intended first birth: Q4 minus Q1	-3.4 (0.6)	-2.7 (0.6)	-2.5 (0.6)	-0.8 (0.7)	-1.3 (0.8)	-6.2 (0.7)	-4.1 (0.8)
Q1 mean	4.1	4.1	3.3	3.3	3.3	8.0	8.0
Enrollment status at t		✓					✓
Highest grade completed at t		✓	✓	✓	✓		✓
1979 expectations, Rotter, parents				✓	✓		
Race and ethnicity					✓		
Woman-years	5,521	5,521	5,055	5,055	5,055	8,617	8,617
Women	2,418	2,418	2,268	2,268	2,268	3,643	3,643

Notes: Top-minus-bottom AFQT quartile difference in the probability of a first birth in the following year, in percentage points, from equation (1) on childless women aged 14–17 (columns 1–5; columns 3–5 enrolled in high school) or 18–20 (columns 6–7). Standard errors clustered by woman in parentheses. Controls are defined in Online Appendix OA.4.

percentage points (s.e. 1.1), against a bottom-quartile rate of 10.4 percent, and -6.8 (s.e. 0.9) for unintended first births. Holding enrollment and grade fixed leaves the gradient where it was (columns 1–2). The 1979 expectations and family background remove the gradient in intended first births altogether (-0.8 , s.e. 0.7, in column 4) but only about a quarter of the gradient in unintended ones, which stays at -5.3 (s.e. 1.0): girls who planned more schooling planned their births later, and family background does not account for the unintended early births of low-ability girls. Adding race and ethnicity as well leaves that gradient at -3.6 (s.e. 1.0) on a bottom-quartile rate of 7.1 percent (column 5). Only at 18–20, when schooling begins to differ, does schooling to date absorb part of the gradient, from -10.3 to -7.0 (columns 6–7).¹¹

¹¹A prospective design in the spirit of Levine and Painter (2003)—the 1,803 girls who were 14–16 and childless in 1979, with only what was measured then as controls—gives the same answer: the gradient in unintended first births by 21 is -22.5 percentage points with age controls alone and -20.1 (s.e. 3.3) with the full 1979 set, while the gradient in intended first births falls from -14.0 to -5.3 (Online Appendix OA.4).

Among adults, whose schooling is largely complete, the gradient also survives within education groups, though it is smaller, and what limits it there is not family background but race. Parents’ schooling and the family’s 1979 income and poverty status leave it where it is; a race indicator removes it.¹² Race is not a confounder to be partialled out here (footnote 15), and the gradient the model is estimated on holds within every race and ethnicity group. Most of the gap opens in the teen years, where education cannot account for it and where, as column 5 of Table 1 shows, the gradient survives family background and race together. These regressions condition on schooling as realized, but schooling is itself a choice, and the opportunity-cost explanation runs through the whole sequence—schooling, work, marriage—not through any one of them with the rest held fixed. Testing it takes a model in which that explanation can re-optimize (Section 5).

Method selection. If high-AFQT women adopted more effective methods, the gradient could reflect the composition of methods rather than per-use performance. They do not: conditional on using a method, low-ability women choose the more effective one. In a multinomial logit on women using reversible contraception, with AFQT quartile, education group, parity and wave fixed effects as regressors,¹³ the predicted probability of using the pill is 76.5 percent in the bottom quartile and 62.5 percent in the top; that of a barrier method is 15.5 percent and 29.4 percent.¹⁴ Low-ability women concentrate on the pill, which fails less often in typical use than a barrier method, and still have more births: method selection works against the ability gradient rather than producing it.

Within-method birth hazard. For each method group $n \in \{\text{pill, barrier, behavioral}\}$, I estimate the linear probability model

$$B_{it} = \alpha_n + \sum_{q=2}^4 \beta_{nq} \mathbf{1}[Q_i = q] + \mathbf{x}'_{it} \gamma_n + \delta_t + \varepsilon_{it}, \quad (2)$$

¹²Online Appendix OA.4 reports the within-education estimates; Online Appendix OA.5.2 adds the two blocks of controls separately and decomposes the movement variable by variable; Online Appendix OA.5.6 estimates the gradient within each race and ethnicity group.

¹³Online Appendix OA.6.2 reports the predicted method-choice probabilities from the multinomial logit.

¹⁴The comparison is among users, not among all women of reproductive age, which is why the gradient runs opposite to NSFG studies that find pill use rising in education (e.g., Sundaram et al., 2017); Online Appendix OA.6.1 reconciles the two (Mosher, 1990; Piccinino and Mosher, 1998).

where B_{it} equals one if woman i had a live birth in the 12-month window following wave t , Q_i is her AFQT quartile (quartile 1 omitted), $\mathbf{x}_{it}$ contains education group, parity (0/1/2+ prior births), age-at-wave indicators, marital status at the previous interview and a window-length indicator,¹⁵ and δ_t are wave fixed effects; the controls are the state variables on which the model’s birth probability depends.¹⁶ Standard errors are clustered by respondent, and a pooled specification adds method-group fixed effects. If higher-ability women have fewer births only because they choose more effective methods, ability should not predict failure within a method, $\beta_{nq} = 0$; if they also use a given method more reliably, $\beta_{nq} < 0$.

The gradient does not disappear within method (panel A of Table 2).¹⁷ In the pooled specification, women in the second, third and fourth AFQT quartiles are 1.3, 1.1 and 1.4 percentage points less likely to have a birth in a year than women in the first ($p < 0.01$, $p < 0.05$ and $p < 0.01$), against a rate of 3.6 percent in the first quartile: the top quartile’s failure rate is about two-fifths lower. Among pill users the top-quartile coefficient is -1.5 points ($p < 0.01$) on a mean rate of 1.7 percent—notable because daily ingestion is a narrow behavioral margin on which cognitive ability can affect consistency within a single technology. Among barrier users the three coefficients are -2.4 , -2.4 and -2.5 pp ($p < 0.10$, $p < 0.10$ and $p < 0.05$), and among behavioral users none is significant.¹⁸

Where ability enters. The accounting used for teen births (Kearney and Levine, 2012; Santelli et al., 2007) writes a woman’s annual chance of a birth as exposure times the share

¹⁵The controls exclude race and ethnicity: AFQT is measured before labor-market entry and absorbs the part of group differences that runs through skills, so conditioning on race would remove part of the variation in ability rather than a confounder (Carneiro and Heckman, 2003; Neal and Johnson, 1996). Online Appendix OA.5.6 reports the estimates within each group. The within-method estimates below are a statement about the pooled population; the structural test of Section 5.3 does not rest on them: the within-method moments carry 14 percent of the test statistic, against 89 percent for the timing and by-quartile wantedness blocks, whose gradients hold within every group.

¹⁶Age enters separately from the wave because birth cohorts span eight years within a wave. Marital status is measured at the interview before the wave, so pregnancy-induced marriage cannot move it. Online Appendix OA.5.1 reports the full estimates, the regressions with the NLSY79 sampling weights and without the age and marital-status controls, and Online Appendix OA.5.2 the regressions that add family background and race separately.

¹⁷Online Appendix Table OA.12 reports the full estimates.

¹⁸Assigning dual users to the most effective tier and within-window method switching both attenuate the gradient toward zero. Across the conventions for classifying a birth without a pregnancy record (Online Appendix OA.5.5) the top-quartile coefficient ranges from -0.014 to -0.022 and is significant under each; unconditional on method it is -3.0 points (Online Appendix OA.5.7).

of the time she uses no method times the non-user birth rate, plus the share she uses one times its failure rate. It separates three margins of birth risk: whether a woman uses a method (use), how often the method she uses fails (failure), and her birth risk when she uses none (fecundity and exposure). Table 2 shows where ability enters. Not through the third margin: among women using no method the birth rate is flat in ability, 2.4 percent a year in the lowest quartile against 2.5 in the highest, with a regression-adjusted difference of +0.4 points (panel B), and intercourse is more frequent at the top, not less.¹⁹ It enters through the first two, the margins of fertility control (panel A): low-ability women are less likely to use any method—62.1 percent of woman—waves in the lowest quartile against 75.9 in the highest—and, when they use one, more likely to see it fail, 3.6 births per hundred user-years against 2.2. The first margin is a choice, over the cost of a method and the value of avoiding a birth; the second is adherence. Non-use is the larger margin in accounting terms (85.1 against 53.0 percent of unintended first births conceived with no method in use, panel C), and the model carries both: the price of a method and the value of a birth govern use, adherence governs failure. Neither margin settles the question on its own: a woman with less to lose has less reason to use a method, so the opportunity-cost account can generate the use gradient. What it cannot generate, once the method is held fixed, is a gradient in how often that method fails, nor—with one adherence level per age—the timing of first births by ability. That is what the test of Section 5 turns on.

2.4 Life-Cycle Facts

The model must also match the broader life-cycle patterns that are its estimation targets (Online Appendix Table OA.3). The gradient is steep early and attenuates with age: at 14–17, 28 percent of women in the lowest quartile have a first birth against 3 percent in the highest; at 18–21, 48 percent against 16 percent; from 22–25 onward the conditional probabilities converge. Mean age at first birth rises from 20.2 to 25.6 across quartiles, 5.5 years, and marriage at first birth from 0.48 to 0.88, yet the share with a child by age 40 falls only from 0.87 to 0.72: the pattern is postponement rather than avoidance of motherhood. Ability also predicts schooling steeply—51 percent of the highest quartile complete college

¹⁹Online Appendix OA.6.4 reports coital frequency by ability quartile and the exposure-margin estimates.

Table 2. Three Margins of Birth Risk by AFQT Quartile

	Q1 level	Q2 difference from Q1	Q3	Q4
<i>A. Fertility control</i>				
Using any method (% of woman–waves)	62.1	6.5 (1.2)	10.7 (1.2)	14.7 (1.3)
Births among users (% per year)	3.6	−1.3 (0.4)	−1.1 (0.4)	−1.4 (0.4)
Pill users	3.1	−1.1 (0.5)	−1.1 (0.5)	−1.5 (0.5)
Barrier users	5.9	−2.4 (1.3)	−2.4 (1.3)	−2.5 (1.3)
Behavioral users	3.9	−0.6 (1.9)	1.5 (2.1)	1.7 (2.3)
<i>B. Fecundity and exposure</i>				
Births among non-users (% per year)	2.4	0.0 (0.6)	0.1 (0.6)	0.4 (0.7)
<i>C. Where unintended first births come from</i>				
Conceived with no method in use (% of unintended first births)	85.1	−5.4 (2.4)	−21.7 (2.9)	−32.1 (3.3)

Notes: The Q1 column reports the lowest quartile’s *raw* level; the Q2–Q4 columns report each quartile’s *regression-adjusted* difference from Q1, so a level and a difference do not add to that quartile’s raw level, with robust standard errors clustered by respondent in parentheses. Panel A: share of woman–waves aged 18–35 using any method, with the quartile coefficients of a linear probability model with the controls of equation (2); the within-method birth rate from the pooled specification of equation (2), and, by method, the quartile coefficients of the same specification estimated within pill, barrier and behavioral users, with the raw Q1 birth rate of each group as the level (full estimates in Online Appendix Table OA.12). Panel B: annual first-birth rate among childless non-users aged 18–33, annualized over the interval to the next wave as in the within-method moments; the differences are regression-adjusted. Panel C: share of unintended first births conceived with no method in use; the differences are raw, with heteroskedasticity-robust standard errors. Sample sizes differ by row, since each comes from its own estimation sample: 25,009 woman–waves for the use margin, 16,087 for the pooled within-method birth rate and 11,299, 3,828 and 960 for pill, barrier and behavioral users, and 3,372 for the non-user rate of Panel B. Online Appendices OA.3.1, OA.6.3 and OA.3 report the full estimates with their samples.

against 3 percent of the lowest—and higher-ability women accumulate more experience at steeper returns, so a career interruption costs them more.²⁰ Premarital childbearing is associated with lower marriage rates and lower spousal earnings (Online Appendix OA.2.3). The wage process, experience and the marriage market must therefore generate ability-graded incentives to delay before any fertility-control channel is invoked.

²⁰The unconditional experience gradient (58 against 26 log points of wage growth over the first five years worked in the highest and lowest quartiles) runs largely through education (Online Appendix OA.10); Online Appendix OA.10.6 scales the within-education gradient up to twice its size and finds that the adherence gradient does not shrink.

3 Model

I develop a dynamic life-cycle model of fertility timing with imperfect fertility control, in the tradition of models of women’s schooling, work, marriage and fertility (Adda et al., 2017; Keane and Wolpin, 2010), with the labor-supply and saving structure of Attanasio et al. (2008) and Ejrnæs and Jørgensen (2020). Its innovation is a fertility-control technology in which women choose among contraceptive methods of known efficacy and cognitive ability shifts adherence—how reliably the chosen method is used.

Costs and benefits of childbearing. Each woman can have at most one child, which saves state space while retaining the margin that drives the ability gradient, the timing of the *first* birth. A birth delivers a “warm glow” from parenthood and utility from child quality, which depends on a one-time monetary investment. It imposes three costs: a *goods cost*, through the household’s consumption-equivalence scale and the investment; an *opportunity cost*, because a birth during the schooling years raises the psychic cost of continuing and thereafter raises the disutility of work, reducing labor supply and slowing experience accumulation; and a *fertility-control cost*, because avoiding a birth requires choosing and using a method, which is costly and imperfectly effective. Assets supply the intertemporal margin under a no-borrowing constraint.²¹

Because ability raises wages and returns to experience, high-ability women face larger opportunity costs of a birth, and education raises the baseline effectiveness of fertility control (Rosenzweig and Schultz, 1989). To these two channels the model adds ability-dependent adherence and an ability-specific intercept in the utility of a child, so that heterogeneous preferences for early motherhood compete with fertility control on equal terms.

Time is discrete and each period is four years. Women enter at age 14 with cognitive ability θ and assets $a_1 = 0$. They remain fertile through age 41, work until 65, and are retired from 66 to 81, receiving Social Security income that depends on education. The decision unit is the household: a single adult before marriage and a two-adult unit that pools income after it; marriage is absorbing. The model abstracts from income uncertainty.

²¹The one-period child could in principle reward delay artificially. Diminishing returns in the child-quality technology, discounting of the utility of parenthood and the finite fertility window (ages 14–41), which makes excessive delay risk involuntary childlessness, limit that force.

3.1 State variables, choices, and timing

The household state at the beginning of period t is

$$\Omega_{it} = \{a_t, \theta_i, e_t, x_t, m_t, k_t, m_k\},$$

where a_t denotes assets; $\theta_i \in \{1, 2, 3, 4\}$ is the cognitive-ability quartile (1 lowest, 4 highest); $e_t \in \{\text{HSD}, \text{HSG}, \text{COL}\}$ is education attainment/status (high-school dropout, high-school graduate, college; HS and Coll in the tables); x_t is accumulated labor-market experience; $m_t \in \{0, 1\}$ is marital status; $k_t \in \{1, 2, 3\}$ records childbearing status (1: no prior birth, 2: first birth occurs in period t , 3: first birth occurred in an earlier period); and $m_k \in \{0, 1\}$ records marital status at the first birth (only relevant when $k_t \neq 1$).

Within a fertile period, single women may meet and accept a partner, childless women choose a contraceptive method n_t and then face stochastic conception, and households choose labor supply l_t , consumption c_t , saving a_{t+1} and, if a birth occurs, child investment i_t . In the two schooling periods the continuation decision—continue high school or drop out at $t = 1$, graduate from college or drop out at the end of $t = 2$ —comes *after* the fertility outcome, so that a birth can raise the cost of staying in school; high-school graduates also choose whether to enter college at the end of $t = 1$. Enrolled women do not marry; college students do not work and receive a student allowance net of the monetary cost of college. After fertility ends the problem is a labor-savings problem, and in retirement labor supply is zero.²²

3.2 Dynamic Household Problem and Value Functions

I present the parts of the problem that matter most for the paper; Online Appendix [OA.7](#) states the full recursion stage by stage.²³

Marriage. Marriage pools resources: spousal income offsets the earnings a mother forgoes, so a married woman values avoiding a birth less than an otherwise identical single woman. If single ($m_t = 0$), the woman meets a potential husband with probability $\mu(e_t, t)$, which

²²Online Appendix [OA.7.1](#) shows the life cycle and the within-period sequencing (Figures [OA.3](#) and [OA.4–OA.5](#)) and gives the complete Bellman equations.

²³Superscripts on the value function (V^1, V^2, V^3) index within-period sub-stages, not time periods.

depends on her education and takes one value within each age group,²⁴ and accepts if the continuation value under marriage is higher. Writing $\Omega_{it}(m)$ for the state with m_t set to m ,

$$V_t^1(\Omega_{it}) = \begin{cases} \mu(e_t, t) \max\{V_t^2(\Omega_{it}(1)), V_t^2(\Omega_{it}(0))\} + (1 - \mu(e_t, t)) V_t^2(\Omega_{it}(0)), & \text{if } m_t = 0, \\ V_t^2(\Omega_{it}), & \text{if } m_t = 1. \end{cases}$$

Contraception and first-birth risk. Only childless women make a fertility-control decision, i.e. when $k_t = 1$ and $t \leq T_F$. They choose a contraceptive method $n_t \in \{0, 1, 2\}$ —no method, a low-efficacy method (barrier or behavioral), or the pill—each of which implies a probability of a first birth in period t , $p_n(\theta_i, g(t), e_t)$, that depends on age, education and, through adherence, ability; Section 3.4 specifies it. Method n carries a utility cost κ_n ($\kappa_0 = 0$) and a Type-I extreme-value taste shock $\sigma_n \varepsilon_{nt}$, and the value of a birth carries a transitory shock $\sigma_\omega \varepsilon_{it}^\omega$, $\varepsilon_{it}^\omega \sim N(0, 1)$ i.i.d. across women and periods, realized before the method is chosen: whether a birth now would be welcome is not fixed by a woman’s circumstances, and ambivalent or shifting desires are common among young women (Edin and Kefalas, 2005; Miller et al., 2013). Only a woman who is sexually active is at risk of conception: with probability $\pi_{\theta t}$, the exposure share, she chooses a method, and otherwise she continues childless. Hence

$$V_t^2(\Omega_{it}) = V_t^{3,j=1}(\Omega_{it}) + \pi_{\theta t} \mathbb{E}_{\varepsilon, \varepsilon^\omega} \max_{n \in \{0, 1, 2\}} \left\{ p_n(\theta_i, g(t), e_t) [\Delta V_t(\Omega_{it}) + \sigma_\omega \varepsilon_{it}^\omega] - \kappa_n + \sigma_n \varepsilon_{nt} \right\},$$

$$\Delta V_t(\Omega_{it}) \equiv V_t^{3,j=2}(\Omega_{it}) - V_t^{3,j=1}(\Omega_{it}),$$

where ΔV_t is the value of a first birth relative to remaining childless and ε_{it}^ω is the idiosyncratic component of that value—a partner, a circumstance, a change of mind—that the survey’s wantedness answer reflects. A woman who wants a birth this period ($\Delta V_t + \sigma_\omega \varepsilon_{it}^\omega \geq 0$) faces no trade-off: using nothing ($n = 0$) carries the highest birth probability at no cost, so she takes it up to the taste shock. A woman who does not chooses n to minimize $p_n |\Delta V_t + \sigma_\omega \varepsilon_{it}^\omega| + \kappa_n$, trading a lower chance of a birth against a higher utility cost. If $k_t \neq 1$, the household skips

²⁴The groups are 18–21, 22–25, 26–33 and 34–41 for women without college, and 22–25, 26–33 and 34–41 for college women, who do not marry while enrolled.

this stage and $V_t^2(\Omega_{it}) = V_t^{3,j=3}(\Omega_{it})$.

Working stage. After the fertility realization the household solves a labor–consumption–saving problem indexed by child status j : $j = 2$ if a first birth occurs in t , $j = 1$ if the woman remains childless, $j = 3$ if she gave birth earlier. The discrete labor choice $l_t \in \{0, 1\}$ carries a Type-I extreme value shock, so $V_t^{3,j}(\Omega_{it}) = \mathbb{E}_\varepsilon \max_{l \in \{0,1\}} \{v_t^{3,j}(\Omega_{it}, l) + \sigma_l \varepsilon_t(l)\}$ with

$$v_t^{3,j}(\Omega_{it}, l) = \max_{a_{t+1}, c_t, i_t \geq 0} \left\{ u(c_t) + [\psi_l(t, e) + \psi_{lk}(e) \mathbf{1}_{\{j=2\}}] \mathbf{1}_{\{l=1\}} + u_k(i_t) \mathbf{1}_{\{j=2\}} \right. \\ \left. + \beta \mathbb{E}[V_{t+1}(\Omega_{i,t+1}) \mid \Omega_{it}, l, a_{t+1}, j] \right\}$$

s.t. $\phi_c(m_t, \mathbf{1}_{\{j=2\}}) c_t + a_{t+1} = (1 + r) a_t + y_t(\Omega_{it}, l) - i_t \mathbf{1}_{\{j=2\}}, \quad x_{t+1} = x_t + \mathbf{1}_{\{l=1\}}.$

Two parameters govern the net utility of work, which is typically negative. The baseline $\psi_l(t, e)$ varies by education and over four life stages, with a retirement term that raises the cost of work at 58–65.²⁵ The second, $\psi_{lk}(e)$, applies only while the child is present and is the child penalty on labor supply (Adda et al., 2017).

College. At the end of $t = 1$, teens who complete high school draw a Type-I extreme value shock and choose whether to enroll at $t = 2$, comparing the beginning-of-period values under each choice, with enrollment carrying a non-pecuniary entry cost $\kappa_C(\theta, k_1)$ that varies with ability and with whether she had a birth at 14–17.²⁶

3.3 Preferences, technologies and transfer system

Preferences over effective consumption. Utility is CRRA over effective consumption, $u(c_t) = c_t^{1-\gamma}/(1-\gamma)$. Household composition enters through the equivalence scale in the budget constraint,

$$\phi_c(m, b) = 1 + \phi_{ca} \mathbf{1}_{\{m=1\}} + \phi_{cb} b,$$

²⁵The stages are ages 14–29, 30–49, 50–53 and 54–65; the retirement term $\psi_{ret} \cdot \max\{t - 11, 0\}$ is common across education groups.

²⁶Online Appendix OA.7.6 states the college-age problem in full.

where $b \in \{0, 1\}$ indicates a child present in the household (a first birth in the period, $j = 2$), so that $\phi_c c_t$ is the expenditure required to attain c_t , with $\phi_{ca} \geq 0$ the additional needs of a second adult and $\phi_{ck} \geq 0$ those of a child present in the household.

Preferences over child quality. If a first birth occurs, parents choose a one-time monetary investment i_t that increases child “quality.” Parental altruism enters as utility from child outcomes:

$$u^k(i_t) = \omega_{0,\theta} + \omega_1 i_t^{\omega_2},$$

where $\omega_{0,\theta}$ is the utility from having a child for a woman of ability θ , so that preferences over motherhood may differ by ability and compete with fertility control in explaining the timing gradient (Section 3.4), ω_1 scales the marginal value of investment, and $\omega_2 \in (0, 1)$ imposes diminishing returns and ensures an interior investment choice.

College psychic cost. College choices are disciplined by a student allowance w_C , the monetary cost of college TC, and an ability-dependent psychic cost. Following the structural education literature, I model the psychic cost of college attendance as

$$\kappa_C(\theta, k_1) = \frac{\xi_C}{\theta^{\omega_C}} + \xi_\theta (\theta - 1) + \mathbf{1}_{\{k_1=2\}} \phi_{k,BAC},$$

where the first term captures the standard finding that psychic costs decline with cognitive ability, the second gives the estimation more freedom in how the cost falls across quartiles (as in [Abbott et al., 2019](#)), and the third is the additional burden of attending college after a teen birth.²⁷ Three further terms make an early birth costly for schooling: a woman with a child pays an additional utility cost to stay in high school ($\phi_{k,HS}$), a teen mother to attend college ($\phi_{k,BAC}$), and a woman who has her first birth while enrolled in college to graduate ($\phi_{k,D}$); graduating also carries a cost ϕ_{grad} for everyone. All four enter utility additively and are restricted to be non-positive, so each cost is the magnitude of its coefficient. The student allowance is $w_C s_\theta$, where w_C is estimated and s_θ scales it by the mean family income of

²⁷Psychic costs of schooling are a standard ingredient in structural models of educational choice (e.g., [Abbott et al., 2019](#); [Keane and Wolpin, 2010](#)); unmeasured monetary costs are observationally equivalent to them.

quartile θ at ages 14–17 relative to the sample mean, measured in the NLSY79 and held fixed.²⁸ The model has no parental sector; the allowance is a reduced-form transfer rule, and indexing it on ability keeps the correlation between family resources and ability from being loaded onto other parameters.²⁹

Progressive taxes and transfers. I adopt the parametric U.S. tax-and-transfer schedule in [Daruich and Fernández \(2024\)](#): gross annual household income $\tilde{y}^0$ (labor earnings, spousal earnings if married, or Social Security in retirement) maps to disposable income $\tilde{y} = \lambda_\tau(\tilde{y}^0)^{1-\tau} + \underline{y}(m_t)$; schooling allowances net of direct costs enter the budget untaxed. $\underline{y}(m_t)$ is a reduced-form lump-sum transfer varying by marital status. Money amounts from the model and the estimated earnings processes are annual, in thousands of 2016 dollars, while a model period covers four years; descriptive tables drawn directly from the NLSY79 state their own units.

3.4 Fertility control technology

Fertility is controlled imperfectly through the choice of a discrete method, whose efficacy is a property of the technology, used with an adherence that is a property of the user. The specification is motivated by the gap between “perfect use” and “typical use” failure rates, which is driven by inconsistent or incorrect use and varies systematically with user characteristics ([Black et al., 2010](#); [Rosenzweig and Schultz, 1989](#); [Trussell, 2011](#)).³⁰

Exposure. Not every childless woman is at risk of conception, and first intercourse comes later for high-ability women. Rather than let adherence absorb this margin, I measure it: $\pi_{\theta t}$ is the share of women in quartile θ who are childless at the start of period t and

²⁸Online Appendix [OA.1.3](#) documents the sample and the construction of every externally measured input.

²⁹The rule follows [Keane and Wolpin \(2001\)](#) and [Johnson \(2013\)](#). Indexing the transfer rather than adding a borrowing constraint follows the evidence that short-run credit constraints explain little of the income gradient in attendance ([Carneiro and Heckman, 2002](#)), while family resources conditional on ability do matter ([Belley and Lochner, 2007](#)). With $s_\theta \equiv 1$ the model reduces to the common-allowance specification.

³⁰Related life-cycle models with imperfect fertility control use a continuous effort mapped through a logit ([Choi, 2017](#)) or a binary choice of whether to try to conceive, with a constant leakage probability otherwise ([Ejrnaes and Jørgensen, 2020](#)). Here the control object is a discrete method with an externally measured efficacy, and ability shifts the reliability with which the method is used rather than the cost of using it or the baseline risk.

have had intercourse by its end, from the NLSY79 age-at-first-intercourse questions.³¹ With probability $1 - \pi_{\theta t}$ the woman is not exposed, conceives with probability zero and uses no method. Treating exposure as data is conservative for adherence: the later first intercourse of high-ability girls is credited to exposure, not to $c_{\theta g}$. The cost is that the counterfactuals hold sexual activity fixed.³²

Methods and adherence. A childless woman of ability θ , education e and age group $g = g(t)$ chooses $n \in \{0, 1, 2\}$: no method, a low-efficacy method (condoms and other barrier methods, withdrawal, rhythm), or the pill. Let p_{be}^0 be the four-year probability of a first birth with no method,

$$p_{be}^0 = \left[p_{\max} \frac{\lambda_{be}}{1 + \lambda_{be}} \right]_{p_{\min}}^{p_{\max}},$$

where $\lambda_{be} > 0$ are the odds that parameterize this *baseline risk*, indexed by education and by an age band b (14–25, 26–33, 34–41), and the truncation keeps the risk in $[p_{\min}, p_{\max}]$.³³

$$p_n(\theta, g, e) = p_{be}^0 [(1 - c_{\theta g}) \rho_n^0 + c_{\theta g} \rho_n], \quad n \in \{1, 2\}, \quad p_0(\theta, g, e) = p_{be}^0. \quad (3)$$

The anchor ρ_n^0 is the counterpart of the typical-use rate, but it is estimated rather than imported, because the NLSY79 failure measure is a birth rate per year of *reported* use and not a first-year typical-use rate.^{34,35} Self-reported use also understates lapses (Potter et al., 1996). Adherence is indexed by the age groups $g \in \{14\text{--}17, 18\text{--}25, 26\text{--}33, 34\text{--}41\}$, with the last set equal to 26–33 because the NLSY79 contraception waves end when the cohort is 35.

³¹Online Appendix OA.1.3 documents how the exposure shares are measured.

³²For a model in which teen sexual activity is a choice, see Di Nola et al. (2025).

³³The odds are ordered by education, $\lambda_{b,HSD} \geq \lambda_{b,HS} \geq \lambda_{b,C}$; the dropout–high-school gap is estimated at 14–25 only. At 14–17, before schooling is resolved, every woman faces the dropout odds and the base cost of the pill. The bands b are coarser than the age groups g that index adherence because no within-method moment exists before 18 to identify a separate teen baseline risk. The pooling shows in the fit: one risk for 14–25 slightly over-predicts first-birth rates at 14–17 and under-predicts them at 18–21 in the three lower quartiles, the direction a lower teen baseline would produce, as would the larger share of teen pregnancies that ended in abortion in these years, 42 percent at 15–17 against 27 percent at 20–24 in 1980–88 (Kost and Henshaw, 2012). The upper bound is a normalization rather than a restriction: the estimated risk is interior in every cell, between 0.73 and 0.75 at 14–25, and the criterion, every baseline risk and every test statistic are the same for any $p_{\max}$ between 0.8 and 1, with the odds rescaling to absorb it; $p_{\min}$ binds in no cell.

³⁴Online Appendix OA.6.1 compares the NLSY79 failure measure with typical-use benchmarks and documents the method mix.

³⁵A missed pill and an unused condom are not the same exposure, so the ratio of pill to low-tier failure among imperfect users is also left to the data (Online Appendix OA.9).

Within each group adherence is parameterized on the logistic scale,

$$c_{\theta g} = \frac{1}{1 + \exp\{-(\alpha_g + \sum_{r \leq \theta} \Delta_{rg})\}}, \quad \Delta_{1g} \equiv 0, \quad \Delta_{rg} \geq 0,$$

so that the intercept α_g sets the bottom-quartile level freely and the cumulative, non-negative increments make $c_{\theta g}$ weakly increasing in ability. The nine increments $\{\Delta_{rg}\}$ are the object of the restriction.

Adherence is disciplined by the within-method birth rates, which are targets of the estimation: equation (3) maps them into $c_{\theta g}$ given the external ρ_n . The sensitivity of the estimates to the data moments (Andrews et al., 2017) assigns 21 percent of the identification of the 18–25 adherence increments to the within-method failure block, second only to first-birth timing at 30 percent (Online Appendix OA.10.5). Two limitations remain. Because the model maps method use into live births, $c_{\theta g}$ absorbs any ability-correlated difference in abortion.³⁶ And because the NLSY79 first asked about methods in 1982, when its youngest respondents were already 18, $c_{\theta,14-17}$ is identified from the teen first-birth hazards and the wantedness of teen first births alone: the same sensitivity calculation puts the within-method block at 6 percent for the teen increments, against 21 percent for the adult ones. The behaviors adherence requires also depend on non-cognitive traits, which measured cognitive ability partly reflects (Almlund et al., 2011), so $c_{\theta g}$ is best read as *fertility-control capacity* rather than cognition alone, which is why Section 6 studies interventions that remove the need for adherence rather than ones that raise skill.³⁷

Method choice. Given the value of a first birth ΔV , which from here on includes the realized transitory component $\sigma_\omega \varepsilon^\omega$, the payoff of method n is $p_n \Delta V - \kappa_n + \sigma_n \varepsilon_n$, with $\kappa_0 = 0$, $\kappa_1 \geq 0$ the utility cost of the low tier, the same for every woman, and $\kappa_2(e) = \kappa_2 + \Delta \kappa_{2,e}$ the cost of the pill. Only the pill carries an education shifter ($\Delta \kappa_{2,HSD} = 0$), unrestricted in sign. It is not an access assumption: it gives education a route into the cost of the more effective method, so that education-specific contraception costs can compete with adherence

³⁶Online Appendix OA.6.5 bounds it: assigning every unreported abortion to the top quartile still leaves a conception gap of 0.104 against a birth gap of 0.417 by age 25.

³⁷Online Appendix OA.9 discusses the interpretation and scope of the fertility-control channel.

as an explanation of the gradient. The choice probabilities are

$$P(n \mid \Omega_{it}) = \frac{\exp\{(p_n \Delta V - \kappa_n)/\sigma_n\}}{\sum_{n'} \exp\{(p_{n'} \Delta V - \kappa_{n'})/\sigma_{n'}\}},$$

and the expected probability of a first birth is $\bar{p} = \sum_n P(n) p_n$. The κ_n are effective costs in units of ΔV , not prices: they stand for everything that makes a method costly to obtain and to use—money, side effects, effort and access—and the estimation infers them from how many women at risk use no method. A policy that changes what a method costs to obtain or to use is a change in κ_n . Because κ_n is the only margin through which the model can explain why women at risk of a birth they report as unwanted use no method, its estimate is large in consumption units. Other structural models of fertility control also specify the cost of a method as a utility term rather than a price (Arcidiacono et al., 2012; Carro and Mira, 2006; Hotz and Miller, 1993), and in the same way it should be read as a residual rather than as the price of a method, which is negligible.³⁸ Choice is where incentives live: within an education–age cell, women of different ability face different ΔV because they face different wages, and they choose different methods for that reason alone.

Unintended births. In the model a first birth is *unintended* if, at the start of the period in which it occurs, the woman would have preferred to remain childless, $\Delta V + \sigma_\omega \varepsilon^\omega < 0$, whether the method she chose failed or she chose none. The second case is the common one: a woman for whom a birth is unwelcome but $|\Delta V|$ is small often chooses no method, and if she conceives she is the survey’s “never used, not wanting” respondent. The definition matches the survey’s,³⁹ and the targeted moment counts women, not births: the share of a quartile’s women whose first birth comes by age 21 and is unintended.

The model nests the opportunity-cost explanation: if $c_{\theta g} = c_g$ for all θ , the ability gradient in first births has to be produced by wages, schooling, marriage and $\omega_{0,\theta}$ alone. Section 5 tests this restriction.

³⁸The residual absorbs forces the model does not represent separately: ambivalence about an early birth beyond what the transitory wantedness component captures (Frost et al., 2012; Kearney and Levine, 2012; Miller et al., 2013), underestimation of the risk of pregnancy and of the efficacy of methods (Frost et al., 2012; Miller et al., 2025), a partner’s disapproval (Delavande, 2008; Miller et al., 2025), and dissatisfaction with side effects, which shows up as inconsistent use (Frost and Darroch, 2008).

³⁹Online Appendix OA.1.7 gives the survey rule.

4 Estimation

Parameters are disciplined in three steps: some are set externally, the earnings process is estimated outside the structural model, and the remaining structural parameters are estimated by the Simulated Method of Moments.

4.1 Externally Set Parameters and Earnings Process

Externally set parameters. The discount factor (0.959 a year) and the coefficient of relative risk aversion (1.98) are [Adda et al.’s \(2017\)](#) estimates. Tuition, taxes and transfers follow published schedules and standard calibrations ([Daruich and Fernández, 2024](#); [Heathcote et al., 2017](#); [National Center for Education Statistics, 2018](#)), method efficacies follow [Trussell \(2011\)](#), and bounds on conception probabilities rule out both perfect control and deterministic fecundity.⁴⁰ The exposure shares $\pi_{\theta t}$ are data: at ages 14–17 they are 0.63, 0.57, 0.46 and 0.36 from the lowest to the highest quartile, 0.82–0.89 at 18–21, and above 0.9 thereafter.

Externally estimated earnings process. Predicted annual real earnings are the model’s deterministic earnings opportunities. For women, $\tilde{w}_t^f = X_t^f \hat{\beta}^f$, where X_t^f contains age and its square, education and ability-quartile indicators, experience, the interactions education $\times$ experience, ability $\times$ experience, education $\times$ ability and education $\times$ ability $\times$ experience; calendar-year effects are absorbed in estimation, so the profiles are those of an average year. Husbands’ earnings, $\tilde{w}_t^h = X_t^h \hat{\beta}^h$, are a reduced-form function of the wife’s age and its square interacted with her education, and of whether she married before her first birth, also by her education, so that assortative mating and marriage selection are captured.⁴¹ Estimating this process outside the structural step ensures that wage–experience profiles by ability are disciplined by labor-market data alone ([Eisenhauer et al., 2015](#)).⁴²

⁴⁰Online Appendix Table [OA.34](#) lists every externally set parameter and its source.

⁴¹Modeling the husband’s earnings as a reduced-form function of the wife’s characteristics, rather than as a separate decision problem, is standard in life-cycle models of women’s choices ([Eckstein et al., 2019](#); [Keane and Wolpin, 2010](#); [van der Klaauw, 1996](#)).

⁴²Online Appendix [OA.10](#) gives the samples and specifications; the estimates are in the Additional Results, Section S.4.

4.2 Estimation and Identification

Estimation. I estimate the remaining parameters by Simulated Method of Moments (SMM) (Gourieroux et al., 1993; McFadden, 1989; Pakes and Pollard, 1989), choosing Θ to minimize the distance between an empirical moment vector $m^{data} \in \mathbb{R}^{138}$ and its model analogue:

$$\hat{\Theta} = \arg \min_{\Theta} (m^{sim}(\Theta) - m^{data})' W (m^{sim}(\Theta) - m^{data}),$$

where W is diagonal and fixed, with $W_{jj} = 1/\hat{\sigma}_j^2$ and $\hat{\sigma}_j$ the bootstrap standard error of data moment j from 1,000 replications resampling women (the two externally measured child-investment ratios, which have none, are weighted as described in Online Appendix OA.11).⁴³ The model is overidentified—138 targeted moments against 76 structural parameters—and standard errors follow from the usual minimum-distance sandwich, inflated for simulation noise (McFadden, 1989).

Moment blocks. The 138 targeted moments⁴⁴ are organized into blocks aligned with the model’s mechanisms: 10 schooling and early-fertility moments, 2 child-investment ratios by mother’s education, 28 first-birth shares by age bin and ability quartile, 17 marriage shares and 36 employment rates by age bin and education, and 25 contraception-use rates by age bin, education and ability half. Three further blocks discipline the fertility-control technology directly: the share of women whose first birth came by age 21 and was reported as unintended, by quartile (4); the unintended share of first births by age at first birth, for the lowest and highest quartiles (8);⁴⁵ and the annual birth rate within pill users and within low-tier users, by quartile (8).⁴⁶ The pill share among users (8) is not targeted.⁴⁷

⁴³I do not use the asymptotically efficient $W = \mathcal{V}^{-1}$: it carries severe small-sample bias (Altonji and Segal, 1996), and $\hat{\mathcal{V}}$ is numerically singular here. Online Appendix OA.11 gives the details, the standard errors, and the distribution theory for the nested test of Section 5.3.

⁴⁴The estimation code carries 146 moments; the eight pill-share moments enter with zero weight and are held out as an untargeted check (Online Appendix OA.12.3).

⁴⁵Online Appendix OA.3 documents the wantedness measures behind both blocks.

⁴⁶The within-method block is compared at an *annual* rate on both sides: the simulated four-year probability p enters the criterion as $1 - (1 - p)^{1/4}$, the convention under which the estimates compare with the annual typical-use rates of Trussell (2011).

⁴⁷The model has no channel for the method mix (Online Appendix OA.6.1), so targeting it would move the method costs without informing the mechanism. The contraception block is measured at the education $\times$ age $\times$ ability level, so that within-cell variation in use is disciplined by the data rather than attributed to adherence by construction.

All parameters are disciplined jointly through the model’s cross-equation restrictions; the blocks are a guide to the main sources of identifying variation, not a claim of one-to-one identification (Adda et al., 2017; Keane and Wolpin, 1997, 2010).

Identification: separating opportunity costs from fertility control. The same ability type that matches schooling, labor supply and marriage must also rationalize fertility timing, as in Keane and Wolpin (2010), so explanatory burden cannot be reallocated across margins without degrading the others. Four restrictions keep within-education birth gradients from being absorbed elsewhere. The earnings process is estimated outside SMM. Schooling costs are disciplined by schooling outcomes, including schooling conditional on early fertility. Exposure $\pi_{\theta t}$ and the allowance scale s_{θ} , the two ability-indexed inputs that are neither preferences nor technology, are measured and held fixed. And baseline risk p_{be}^0 , the probability of a first birth with no method, is common within age–education cells, so within-cell ability gradients load on adherence, preferences or method choice.⁴⁸ The last restriction is what lets the within-method birth rates identify adherence, and it is a restriction the data support rather than an assumption of convenience. Baseline risk is the chance of a first birth when no method is used, which depends on fecundity and on how often a woman has intercourse, not on which method she chooses: method choice is already held fixed when failure is compared within a method. Had baseline risk been allowed to vary with ability, equation (3) could not have told the two apart: pill users of high ability could have fewer births either because they take the pill more reliably or because they would conceive less often even without it. Women who use no method rule the second channel out: for them the birth rate is the baseline risk itself, so an ability gradient in baseline risk would appear there. It does not. Among childless women using *no* method the first-birth rate over the interval to the next wave, 1.8 years on average, is 4.2 percent in the lowest quartile and 4.7 percent in the highest (2.4 and 2.5 percent a year, Table 2), and the top-quartile gradient within education, age and wave is 0.0071 (s.e. 0.0122), slightly *positive*, where a baseline-risk explanation requires it to be negative.⁴⁹ The check has limited power: the non-user cells are

⁴⁸Method choice is *not* restricted to be constant across ability groups within an education–age cell: the choice probabilities depend on $-\Delta V$, which varies with ability through wages and continuation values, and the within-cell variation in method use by ability is itself targeted (25 moments).

⁴⁹Online Appendix OA.6.3 reports the non-user birth-rate estimates.

small, and the 95 percent interval around that gradient, about -1.7 to $+3.1$ points, does not exclude a baseline-risk gradient as large as the within-method one. Section 5.4 therefore estimates the alternative directly, letting ability move baseline risk instead of adherence, and both together. Coital frequency runs the same way: intercourse is *more* frequent at higher ability, by 1.92 acts a month between the bottom and top quartiles (s.e. 0.44), so conditioning on it moves the within-method gradient from -0.0224 to -0.0239 , away from zero.⁵⁰

Identifying fertility-control parameters. Each object of the technology has a block of moments that speaks to it most directly.⁵¹ Baseline risk p_{be}^0 (through the odds λ_{be} that parameterize it) is disciplined by the level and age profile of first births, given the education composition of each quartile that the schooling block pins down. Holding the method fixed, equation (3) makes the ability gradient in within-method failure a gradient in $c_{\theta g}$, while the level of failure and its pill-to-low-tier ratio, common across quartiles, pin ρ_n^0 . The method costs κ_1, κ_2 and the dispersion σ_n are disciplined by the share using any method by education, age and ability half, given the ΔV the rest of the model implies. The preference intercepts $\omega_{0,\theta}$ are disciplined by the unintended-birth moments: given how many women have a first birth by 21, which the timing moments pin down, the block says how many of those births came to women who would rather have waited. Raising $\omega_{0,\theta}$ turns unintended births into intended ones at given adherence and method costs, while adherence and method costs move the share only through births to women trying to avoid one. The scale of the transitory component is disciplined by the composition of first births by age at first birth: the preference intercepts move the quartiles' *levels*, while σ_ω moves how far, within a quartile and an age, the classification departs from all-or-nothing.⁵²

Consider two childless women of the same education and age but different ability. If both use the pill, their birth rates differ only through adherence, and the within-pill moment measures that difference directly. If they choose their method, the higher-ability woman's larger $-\Delta V$ makes her more likely to use one, and to use the pill—an ability gradient in

⁵⁰Online Appendix OA.6.4 reports the coital-frequency estimates.

⁵¹Online Appendix Table OA.33 maps the objects to the moments and the variation used.

⁵²Online Appendix OA.3 reports the scale of the shock against the dispersion of ΔV that the state produces.

use and in births with no difference in adherence at all. The any-method moments discipline how strongly choice responds to ΔV , and the unintended-birth moments discipline $\omega_{0,\theta}$, so the estimator cannot attribute to adherence a gradient that incentives or preferences produce. Each of the three therefore draws its identifying variation predominantly from a different block—adherence from within-method failure, incentives from method use, preferences from wantedness—though first-birth timing contributes to all three, which is why Online Appendix [OA.10.5](#) reports the shares rather than asserting exclusion.

5 Results

Section [5.1](#) shows the fit, Section [5.2](#) the estimated fertility-control technology, and Section [5.3](#) the test of whether opportunity costs and preferences alone can account for the data.

5.1 Model fit

The model fits the data well. Figure [2](#) compares it with the NLSY79 on the four outcomes at the center of the paper’s argument—the timing of first births by ability quartile, the share of early births that were unintended, and the two margins of fertility control, use and failure—and it reproduces the level and the ability gradient of each. It also matches the other targets: schooling by ability, child investment, and the life-cycle profiles of employment, marriage and contraceptive use by education.^{[53](#)}

The model captures the ability gradient in the timing of first births and its narrowing with age (Figure [2a](#)). By age 21, 58 percent of the lowest quartile and 20 percent of the highest have had a first birth in the model, compared with 63 and 18 percent in the data, and by age 41 the gap is 17 percentage points in the model and 15 in the data. 27 of the 28 timing moments are within five percentage points of their targets. The model matches the ability gradient in unintended first births almost exactly (Figure [2b](#)): the share of women with an unintended first birth by 21 falls from 31.9 percent in the lowest quartile to 10.3

⁵³Online Appendix [OA.12](#) plots these profiles and reports all 138 moments with their contribution to the criterion.

percent in the highest, compared with 32.5 and 10.2 percent in the data.

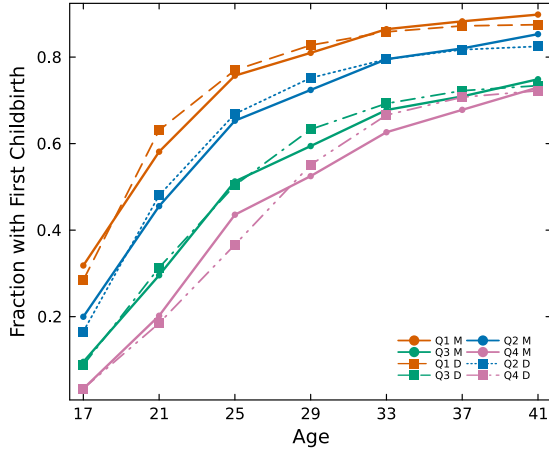
The model captures the two margins of fertility control of Section 2.3. Use of any method rises with ability in the model as in the data, from 56 to 65 percent of childless women between the lowest and the highest quartile against 47 to 72 percent (Figure 2c). Within method, the annual first-birth rate declines with ability for both the pill and the low-tier methods, as in the data (Figure 2d); these are the moments that identify adherence. The third margin of birth risk, fecundity and exposure, is not estimated by ability: exposure is set at its measured value in each quartile, and the birth rate of a woman using no method is common within education and age. This is consistent with the evidence of Section 2.3 that ability does not enter through this margin.⁵⁴

Two features of the fit deserve comment. First, the model places a few more high-ability first births in the early twenties than the data: 44 percent of the highest quartile have had a first birth by 25 in the model against 37 percent in the data, with correspondingly fewer between 26 and 33. This reflects the transitory wantedness shock, whose dispersion is common across quartiles; it does not change the ordering of the quartiles at any age. Second, pill failure in the two lowest quartiles is somewhat higher in the model than in the data, 3.1 and 1.5 percent a year against 2.0 and 1.0 percent, a difference of about one percentage point, and up to two for the low-tier methods, whose failure the model over-predicts in every quartile; both are the price of matching the lowest quartile’s early births. Among the moments not shown, the largest miss is contraceptive use among high-school dropouts at ages 22–25 (49 percent in the model against 30 in the data). The cost of each method is common across education groups, and for a childless dropout the value of delaying a birth is small, so her use is governed by costs that are set to fit use in all groups; a common cost cannot match the dropouts’ low use without lowering use elsewhere. The miss does not carry over to births: the model matches these women’s first births, and the ability gradient is identified within education.

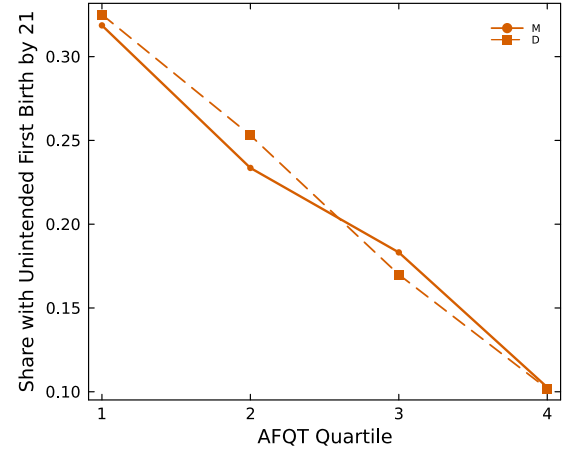
As an untargeted check, if the ability gradient reflected which method women choose

⁵⁴Use by quartile is not targeted; the estimation targets use by education and ability half, which the model matches (Online Appendix Figure OA.6b). The model produces the by-21 gap in unintended births mostly through method failure rather than non-use, the reverse of the data’s route in panel C of Table 2; Online Appendix OA.12.4 reports the split and treats it as a limitation.

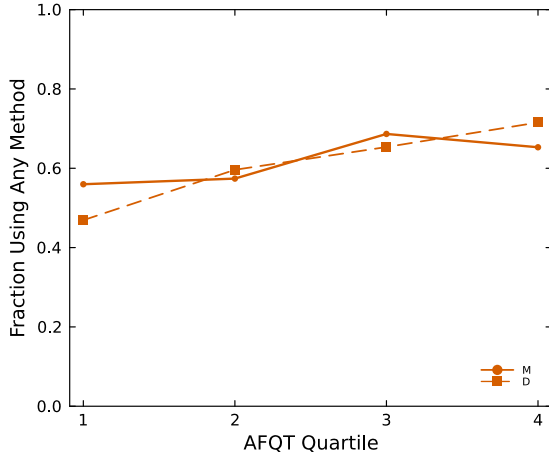
Figure 2. Model Fit: First-Birth Timing, Wantedness, and Fertility Control



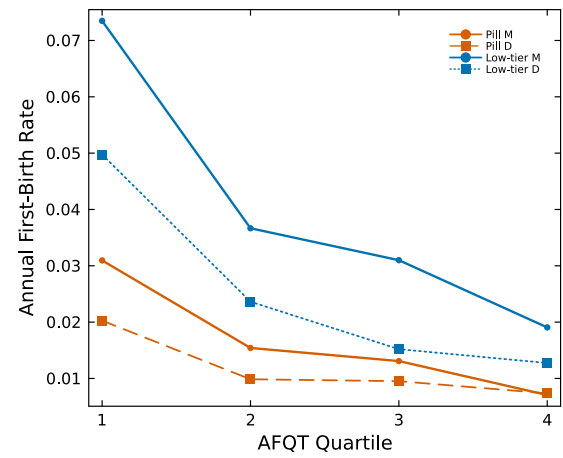
(a) First birth by age, by ability quartile



(b) Unintended first birth by 21, share of women



(c) Use of any method, childless women 18–33



(d) Annual first-birth rate within method, ages 18–33

Notes: Panel (a): cumulative fraction of women with a first birth by the age shown, by AFQT quartile (Q1 = lowest). Panel (b): share of women in each quartile with a first birth by 21 that they reported as mistimed or unwanted. Panel (c): share of childless women at risk who use any method, ages 18–33, by quartile. Panel (d): annual first-birth rate among childless women using the pill and among those using a low-tier method, ages 18–33, by quartile. Solid lines with circular markers are the model, dashed lines with square markers the NLSY79 data.

rather than how consistently they use it, the pill share among users would rise steeply with ability. It does not: among childless women aged 18–33 it is flat in the data, 0.835 in the lowest quartile and 0.824 in the highest, and the model, which places no weight on these moments, predicts it flat as well.⁵⁵ The model does not manufacture a gradient in method choice where the data have none.⁵⁶

5.2 The Estimated Fertility-Control Technology

Table 3 reports the estimated adherence parameters $c_{\theta g}$ by ability quartile and age group, with standard errors, together with the differences from the lowest quartile (panel C).⁵⁷

Adherence rises steeply with ability among teenagers and young adults and converges across quartiles by the late twenties. At ages 14–17, adherence is 0.25 in the lowest quartile, 0.59 in the second and 0.80 in the third; at 18–25 it rises from 0.54 in the lowest quartile to 0.88 in the highest. These levels are the ones the adherence literature measures directly: 47 percent of pill users miss at least one pill per cycle (Rosenberg et al., 1998), women aged 18–20 followed weekly use contraception consistently in 72 percent of weeks (Stidham Hall et al., 2013), and 40 percent are always consistent against 11 percent never (Wu et al., 2016). By 26–33 adherence is essentially complete in every quartile (0.99 to 1.00), and the differences between quartiles are at most 0.008 and carry no standard error. The age profile is also the one that literature finds: failure of user-dependent methods is nearly twice as high under 21 as above, while the failure of long-acting methods does not vary with age (Winner et al., 2012). Since perfect-use failure is close to zero, the ratio of non-adherence, $(1 - c_{Q1,g})/(1 - c_{\theta g})$, is the ratio of the failure risks two users of the same method face: a pill user in the lowest quartile faces about 3.8 times the failure risk of a user in the highest

⁵⁵This is the raw pill share among method users in a narrower sample—childless women aged 18–33—and is not the multinomial-logit prediction of Section 2.3, which conditions on all reversible-method users at a wave and holds education, parity and wave fixed. The two are not comparable, and only the first is a model moment.

⁵⁶The model predicts a lower level of the pill share than the data (0.482 to 0.547 across quartiles), because the estimation weights the moments that identify adherence—unintended births and within-method failure—over the method mix. Re-estimating the method costs to match the pill share does not change the conclusions: the fall in first births by 21 from equalizing adherence is 13.4 rather than 16.1 percent, the fall from the long-acting method at Colorado’s take-up 6.9 rather than 7.1 percent, and the lowest quartile’s welfare gain from that method 10.2 rather than 9.9 percent of consumption (Online Appendix OA.12.3).

⁵⁷Online Appendix OA.13 reports the full parameter vector for the estimated and the restricted model and plots adherence against the restricted model (Figure OA.7).

at ages 18–25, and about 3.8 times that of a third-quartile user at 14–17.⁵⁸

The ability differences are precisely estimated where they matter. The gap between the lowest and the third quartile at 14–17 is 0.55 (standard error 0.17) and the gap between the lowest and the highest quartile at 18–25 is 0.34 (0.09); the second- and third-quartile differences are significant at the one percent level at both ages, as is the fourth-quartile difference at 18–25, and the second-quartile differences are about two-thirds the size of the larger ones; the fourth-quartile difference at 14–17 carries no standard error because adherence in that cell is nearly saturated. The narrowing with age is estimated as sharply: the gap between the lowest and the highest quartile at 18–25 exceeds the one at 26–33 by 0.33 (0.09). The estimates establish a gradient in fertility control between the bottom quartile and the top half that is concentrated before 26—the years in which an unintended birth is most costly and in which the ability gradient in first births opens up.

Could the adherence gradient instead reflect a misspecified earnings process, with a steeper wage gradient doing the work? I re-estimate the fertility-control parameters with the ability gradient in earnings scaled up by a factor of up to two, well beyond what measurement error in the AFQT could justify. The adherence estimates are robust: at one and a half times the estimated gradient they move by less than five percent, and at twice it the teenage gap falls by 1 percent and the young-adult gap not at all, while the fit of the model worsens. The reason is that adherence is identified by the within-method birth rates, which condition on the method in use and do not respond to wages.⁵⁹

⁵⁸Adherence for the highest quartile at 14–17 is estimated at 0.998, where the logistic is nearly saturated, because near-perfect use is the only margin on which the model can bring these women’s birth rate at 14–17 down to the 3.4 percent in the data; the criterion is flat in its logit, so no standard error is defined and the teen gradient is read off the third quartile. Six parameters carry no standard error: the 26–33 intercept and its second- and third-quartile increments, where adherence is essentially complete in every quartile and the criterion is flat in them; the top-quartile increment at 26–33, which sits on the upper bound of its box; the saturated top-quartile increment at 14–17; and the dropout increment to the baseline-risk odds at 14–25. Online Appendix [OA.13](#) lists them. The utility cost of the low-tier method, κ_1 , is estimated at 0.016 and is not distinguishable from zero, so the pill’s cost κ_2 is in effect identified relative to a free alternative.

⁵⁹Online Appendix [OA.10.6](#) reports the sensitivity analysis.

Table 3. Estimated adherence $c_{\theta g}$, by AFQT quartile and age group

Ages	Q1 (lowest)	Q2	Q3	Q4 (highest)
<i>A. Estimated model: adherence $c_{\theta g}$</i>				
14–17	0.250 (0.203)	0.588 (0.154)	0.801 (0.079)	0.998 [‡]
18–25	0.544 (0.123)	0.761 (0.079)	0.782 (0.064)	0.879 (0.043)
26–33	0.992 [‡]	0.994 [‡]	0.999 [‡]	1.000 [‡]
<i>B. Restricted model: $c_{\theta g} = c_g$</i>				
14–17	1.000 [†]	1.000 [†]	1.000 [†]	1.000 [†]
18–25	0.692 (0.157)	0.692 (0.157)	0.692 (0.157)	0.692 (0.157)
26–33	1.000 [†]	1.000 [†]	1.000 [†]	1.000 [†]
<i>C. Ability gaps, estimated model: $c_{\theta g} - c_{Q1,g}$</i>				
14–17	–	0.337*** (0.115)	0.551*** (0.174)	0.748 [‡]
18–25	–	0.217*** (0.070)	0.238*** (0.077)	0.335*** (0.094)
26–33	–	0.002 [‡]	0.007 [‡]	0.008 [‡]

Notes: $c_{\theta g}$ is the probability that a woman of AFQT quartile θ in age group g uses her chosen method as intended. Panel A: estimated model; Panel B: the restricted model with $c_{\theta g} = c_g$ (Section 5.3); Panel C: the gap relative to Q1. Standard errors in parentheses, from the sandwich of Section 4 by the delta method; in Panel C they are the standard errors of the difference, which account for the correlation between cells. Adherence at ages 34–41 is set equal to its 26–33 value and not estimated, because the NLSY79 contraception waves end when the cohort is 35. ***, ** and *: significant at the 1, 5 and 10 percent levels. [‡] marks a cell that contains a parameter in which the criterion is flat—the logistic is saturated near one, or the increment is not separately identified—so no meaningful standard error can be computed: in Panel A Q4 at ages 14–17, Q1 at ages 26–33, Q2 at ages 26–33 and Q3 at ages 26–33. [†] marks a parameter estimated on a bound of its box, where the estimator is not asymptotically normal and no standard error is reported: in Panel A Q4 at ages 26–33; in Panel B the common level at ages 14–17 and ages 26–33, whose intercept sits on the upper bound and marks the whole row.

5.3 Nested Specification Test: Is Ability-Dependent Fertility Control Necessary?

The test asks whether opportunity costs and preferences alone can account for the ability gradient in first births. The restricted model imposes

$$H_0 : c_{\theta g} = c_g \quad \text{for all } \theta,$$

by fixing the nine adherence increments at zero: adherence may vary with age but not with ability. Every other parameter is re-estimated on the same 138 moments, weights and bounds—schooling costs, marriage, the method costs κ_n and their dispersion σ_n , the non-adherent failure rates ρ_n^0 , the scale of the transitory wantedness component σ_ω , and, most importantly, the ability gradient in preferences $\omega_{0,\theta}$. The restricted model is therefore the best model without the adherence channel that the data admit, not the estimated model with the mechanism switched off; in particular it remains free to reproduce the ability gradient in whether a method is used, since method choice responds to $-\Delta V$ and to the costs κ_n .

The minimized criterion is 458.5 for the estimated model and 556.0 for the restricted one. Because the models are nested, the difference $D = L_R - L_U = 97.5$ is a test statistic for the nine restrictions, and under the null it is asymptotically distributed as χ_9^2 , whose one-percent critical value is 21.7. The hypothesis of common adherence is rejected: the statistic is 4.5 times the critical value.⁶⁰

Where the restricted model fails. Table 4 breaks the increase in the criterion down by block of moments. Most of it, 81 percent, comes from the timing of first births. Without ability differences in adherence, the model cannot keep teen births concentrated at the bottom of the ability distribution: the share of the highest quartile with a first birth by 17 rises to 4.6 percent, against 3.4 in the data, and the lowest quartile’s share falls from 8.4 times the highest’s in the data (9.5 in the estimated model) to 7.1. Failure within a method adds 14 percent: it declines with ability in the data and in the estimated model, and is essentially flat under common adherence.⁶¹ Wantedness, by contrast, is not where the restricted model fails. It reproduces most of the ability gradient in unintended first births by 21, 18 percentage points against 22 in the data, and the two wantedness blocks account

⁶⁰The χ_9^2 distribution assumes the optimal weighting matrix and a null in the interior of the parameter space; neither holds exactly here, and neither affects the conclusion. With the diagonal weighting matrix the statistic is a weighted sum of chi-squares, and the mean-matched version that corrects for this is $D_{NW} = 87$; because the null fixes the adherence increments at the boundary of their parameter space, the asymptotic p -value is conservative. The two wantedness blocks together carry 5 percent of the increase in the criterion, and the rejection survives their removal: re-estimating both models on the 126 moments that exclude them gives $D = 141.1$, 6.5 times the one-percent critical value; the unrestricted model’s search on that moment set was still improving when it reached its time limit, so this D is a lower bound. Online Appendix OA.11.4 gives the distribution theory and reports 5 further starts of the restricted model drawn around the reported estimate; the unrestricted model is not multi-started, which makes D conservative.

⁶¹Online Appendix Table OA.36 reports both models on each of these moments.

for only 5 percent of the increase. It also fits contraceptive use better than the estimated model.

The parameters show how it compensates. With one adherence level per age, the restricted model can keep high-ability women from early births only through preferences and use: it makes a child less attractive to the highest quartile (the increment $\omega_{0,Q4} - \omega_{0,Q1}$ falls from -0.07 to -0.22), raises the dispersion of the transitory wantedness component (from 0.212 to 0.287), re-sets the method costs, and pushes common adherence to the upper bound of its range at 14–17 and 26–33. It also raises the career cost of a child, which worsens the fit of the employment profile (36 percent of the increase). Even with all of these margins free, a model without ability differences in adherence cannot reproduce the timing of first births by ability or the decline in failure within a method, and this is why the mechanism is necessary.

Is the opportunity-cost channel too weak by construction? A woman in the model has at most one child, present for a single four-year period, so the model cannot produce the multi-year career interruptions that [Adda et al. \(2017\)](#) measure. If this made the opportunity cost of an early birth too small, the restricted model would lose for the wrong reason. Three results indicate that it does not. First, an early birth still has lasting costs: a period out of the labor force is a permanent loss of experience, whose return rises with education and ability. What the model leaves out is the time cost of an older child, not the persistence of the wage loss. Second, the restricted model already leans on this channel: the child penalty on labor supply, ψ_{lk} , rises from -0.061 to -0.714 for high-school dropouts and from -0.021 to -0.053 for high-school graduates,⁶² and its criterion is still 21 percent higher than the estimated model’s. Third, a larger opportunity cost acts on the wrong margin. It changes whether a woman uses a method, not how often the method fails once she does, and timing and failure are where the restricted model fails. Doubling the ability gradient in earnings, which raises the earnings gap between the highest and lowest quartiles for a high-school graduate with ten years of experience to \$14.3k, leaves the estimated adherence

⁶²Neither estimate is at a bound: the dropout penalty lies 51.4 percent above the lower end of its admissible range. The criterion is flat in it at the restricted estimate, so its level is loosely determined; the direction of the change, which the high-school penalty shares, is what matters here.

gaps essentially unchanged and raises the criterion by 9 percent.⁶³

Could education, rather than ability, shift adherence? The restriction removes ability from adherence without handing it to education, so the test on its own shows that adherence must vary across women, not that it must vary with ability. Three features of the evidence point to ability. First, education already enters the fertility-control technology in both models, through the baseline risk p_{be}^0 and through the education shifter in the cost of the pill, whose sign is free: an education gradient in how often women conceive or in what the more effective method costs is available to the restricted model, and does not rescue it. Second, the gap the restricted model misses opens before schooling is resolved. More than half of the gap in first births by 21 is in place by 17, while girls in every quartile are still in school, and in the data the gradient survives schooling measured before the conception (Section 2.3). An adherence shifter indexed by completed education cannot act at 14–17, which is where the restricted model places too many first births in the highest quartile. Third, the within-method gradient is a gradient within education: the hazard regressions of Section 2.3 condition on education group, whereas a channel that reached within-method failure only through the education composition of each quartile would leave no gradient there.

5.4 Adherence or Baseline Risk?

The identifying restriction of Section 4.2 is that baseline risk, the chance of a birth when no method is used, does not vary with ability within age–education cells. The evidence on non-users supports it, but cannot rule out an ability gradient in baseline risk as large as the one within methods. I therefore estimate two further models. In the *baseline-risk* model, ability shifts baseline risk instead of adherence: baseline risk varies by quartile through nine increments, as many as the adherence increments, and adherence is common across quartiles. In the *encompassing* model, ability shifts both baseline risk and adherence, so it contains the estimated and the baseline-risk models as special cases. The estimated and baseline-risk models cannot be tested against each other directly, because neither is a special case of

⁶³Online Appendix OA.10.6 reports this exercise and intermediate values of the scaling. It changes the size of the opportunity cost, not its duration.

Table 4. Nested specification test: the increase in the criterion under common adherence, by block of moments

Moment block	# moments	Criterion		Share of increase
		Estimated	Restricted	
First births by age $\times$ quartile	28	137	216	81%
Unintended first birth by 21	4	5	12	8%
Unintended share by age at first birth	8	30	27	−3%
Contraception use	25	84	54	−31%
Employment	36	123	158	36%
Within-method failure	8	28	42	14%
Marriage	17	13	8	−5%
Schooling	10	38	33	−5%
Child investment	2	1	5	4%
Total	138	458.5	556.0	100%

Notes: Weighted criterion by block of moments for the estimated model and for the restricted model with $c_{\theta g} = c_g$; the last column is each block’s share of the increase in the total, $D = L_R - L_U = 97.5$, which is the test statistic of Section 5.3 (nine restrictions; χ^2_9 one-percent critical value 21.7). Both models use the same 138 moments, weights and bounds; the restricted model fixes the nine adherence increments at zero and re-estimates every other parameter. Shares sum to 100 up to rounding; a negative share is a block the restricted model fits better, the largest being contraceptive use, which it fits by raising method costs (panel D of Online Appendix Table OA.36).

the other, but each can be tested against the encompassing model: removing its baseline-risk increments asks whether ability in baseline risk adds anything once adherence varies with ability, and removing its adherence increments asks the reverse. All four models are estimated on the same 138 moments, weights and bounds.⁶⁴

Table 5 reports the four criteria and four tests, the second of them the test of Section 5.3. Against the restricted model, ability in baseline risk lowers the criterion by 2.2 percent, and ability in adherence by 17.5 percent. Within the encompassing model, the baseline-risk increments lower the criterion by a further 0.4 percent ($p = 0.995$) in a model in which adherence already varies, and the adherence increments lower it by 16.0 percent ($p = 0.000$)

⁶⁴The reported estimate of each model is the better of its own search and the estimates of the models it contains, which also serve as starting points for the encompassing model, so every difference in the criterion is non-negative by construction.

Table 5. Adherence, baseline risk, or both

Model	Ability moves		Parameters	Criterion
	Adherence	Baseline risk		
Restricted (Section 5.3)			67	556.0
Baseline risk		✓	76	543.9
Estimated	✓		76	458.5
Encompassing	✓	✓	85	456.8
Test	D	Restrictions	p	D_{NW}
Baseline risk vs. restricted	12.1	9	0.206	10.7
Adherence vs. restricted	97.5	9	0.000	87.2
Baseline risk, given adherence	1.7	9	0.995	2.0
Adherence, given baseline risk	87.1	9	0.000	97.8

Notes: Minimized criterion on the 138 targeted moments. D is the increase in the criterion from the restriction in each row; p is from a χ^2 with as many degrees of freedom as restrictions, conservative because the null lies on the boundary of the parameter space; D_{NW} is the mean-matched statistic of Online Appendix OA.11. Each unrestricted model's estimate is the better of its own search and the models it nests.

in one in which baseline risk already varies. The pattern is one-sided. Given the same nine parameters, ability in baseline risk lowers the restricted model's criterion by only 2.2 percent, and its criterion is 19 percent higher than with ability in adherence; within the encompassing model the baseline-risk increments add little once adherence varies, while the adherence increments still add a large amount once baseline risk varies. The within-method gradient is read as adherence, then, not because the identifying restriction assumes it but because the alternative channel, given the same freedom, does not fit it. The baseline-risk model also cannot use the adherence it is left with: its common adherence is at the ceiling in every cell at 14–17 and 26–33 and is 0.51 at 18–25, with a standard error of 0.41, so with ability in baseline risk the within-method moments no longer pin adherence at all.

A second concern is that the restricted model is allowed only a mean shift in preferences by ability, through $\omega_{0,\theta}$, while the estimated model gets nine adherence increments. Letting the dispersion of the transitory wantedness component vary by quartile as well, in both models, leaves the restricted model's criterion 23 percent above the estimated model's, against 21

percent without it (Online Appendix [OA.11.5](#)). The dispersion channel improves the fit of both models but closes none of the gap between them, and the estimated dispersion is not systematically smaller at high ability.

6 Counterfactual Experiments and Policy Implications

In this cohort, and within this model, low-ability women have more early, unintended births because they use contraception less reliably, not only because they use it less or value children differently. If the gradient is in adherence, policies that lower the cost of contraception and policies that remove the need for adherence differ not only in how much they reduce early births but in whom they reach. Three experiments quantify the difference. The first gives every woman the adherence of the highest ability quartile at her age, holding method costs, preferences and wages at their estimates; it is an upper bound on what counseling or adherence support could deliver. The second removes the cost of using either method for every woman. Because κ_n is the full cost of use rather than a price, this is an upper bound on what any policy acting on the cost of contraception can do.⁶⁵ The third adds a long-acting reversible method whose failure rate does not depend on adherence, at the utility cost that reproduces the take-up of Colorado’s Family Planning Initiative, 25 percent of users at ages 18–25. This is the policy of the programs that offered intrauterine devices and implants through public clinics ([Lindo and Packham, 2017](#); [Secura et al., 2014](#)).⁶⁶

⁶⁵The Affordable Care Act’s contraceptive mandate and the Medicaid family-planning expansions studied by [Kearney and Levine \(2009\)](#) lowered out-of-pocket prices, a small part of that cost. Formally, $\kappa_n \mapsto (1 - \delta)\kappa_n$ with $\delta = 1$; the pill’s education shifter is additive and not scaled. Removing the cost of use lowers first births by 21 by 14 percent, against 16 percent for the adherence experiment. This ranking is not sharp—it holds in 60.0 percent of the 200 bootstrap draws—but who gains is: the lowest quartile gains least from removing the cost of use in 99.0 percent of draws and most from equalized adherence in 91.5 percent (Online Appendix [OA.16](#)). A partial cut, $\delta = 0.72$, which lowers first births by 21 by ten percent, ranks the quartiles’ gains the same way (Online Appendix [OA.15](#)).

⁶⁶The long-acting method’s first-birth risk is $p_{be}^0 \rho_L$ with $\rho_L = 0.005/0.85$, the midpoint of the 0.2–0.8 percent IUD failure range in [Trussell \(2011\)](#) over the no-method rate. That range runs from perfect use of the levonorgestrel IUD to typical use of the copper IUD, so ρ_L sits above the perfect-use basis of ρ_1 and ρ_2 ; this is conservative, as is leaving out implants, whose failure rate is lower still. It shares one taste draw with the pill, so the most birth-averse pill users switch first, as in Colorado, where the rise in long-acting use was mirrored by a fall in pill use. Two variants bound the technology channel: the method at the pill’s cost, and every method in use made adherence-proof. Because adherence at 14–17 is identified from the teen birth and wantedness moments alone and is saturated for the top quartile, every experiment holds each quartile’s 14–17 adherence at its 18–25 value, which the within-method moments identify, and the baseline is simulated under the same restriction; the unrestricted version raises the magnitudes of the adherence

Fertility. Table 6 reports the effects of the three experiments. Closing the adherence gap lowers the share of women with a first birth by age 21 from 0.370 to 0.310, a fall of 16 percent. Removing the cost of use lowers it to 0.318, a fall of 14 percent, about as much as the adherence experiment; a long-acting method at Colorado’s take-up lowers it to 0.344. The two main policies also reach opposite ends of the ability distribution. Closing the adherence gap acts on the bottom: first births by 21 in the lowest quartile fall from 0.536 to 0.400, while the highest quartile is untouched by construction. Removing the cost of use does the reverse, lowering first births by 21 by 9 percent in the lowest quartile and by 20 percent in the highest: lower costs are taken up most by the women who already controlled their fertility well. The reason is that more use is not more effective use. Removing the cost of use raises the share of childless women using any method at ages 18–25 from 0.741 to 0.805, far more than closing the adherence gap does (0.746), and the pill’s share among users rises from 0.556 to 0.617 because the cut is largest for the costlier method, but the marginal users are the poorest adherers, so the share of first births that are unintended does not fall: it moves from 0.392 to 0.412, whereas closing the adherence gap lowers it to 0.334.

Welfare. Table 7 reports the welfare gains as consumption-equivalent variation, the permanent proportional change in consumption that gives a woman at age 14, before any shock is drawn, the value she attains under the experiment.⁶⁷ Two features govern how it should be read. First, the utility flow from a child is an estimated parameter and the measure counts the mother’s utility only, so the numbers say how the gains from each experiment are distributed across ability quartiles—who gains and who does not—rather than how much any experiment is worth in absolute terms. Second, removing the cost of use benefits a woman in two ways. Every woman already using a method stops bearing the cost she was bearing anyway, a windfall she collects without changing any decision; and some women respond by using a method, or a better one, and avoid births they did not want. Only the second is the value of the policy as fertility control, and only it is comparable with closing

experiments by between a half and a factor of two, leaves the cost cut where it is, and leaves the comparison between instruments unchanged.

⁶⁷The measure is ex ante and integrates over every taste shock, so it credits a new option only where it is chosen for its payoff. The conversion into consumption units values a proportional change in consumption at the baseline choices.

the adherence gap, which changes no cost. The lower panel of Table 7 separates the two. The windfall to existing users is 87 percent of the lowest quartile’s total gain and 74 percent of the highest’s. Its level is not that of a subsidy that lowers the price of contraception by a few dollars: the experiment removes the whole of κ_n , which also stands for ambivalence about the birth, misperceived risk and a partner’s resistance (Section 3.4). The consumption-equivalent gains from this experiment are therefore an upper bound, and what they establish is who gains. The windfall is smallest for the lowest quartile, because high-ability women use methods, and the pill, more, and net of it the highest quartile still gains more than the lowest (Table 7). Closing the adherence gap, by comparison, is worth 29.7 percent to the lowest quartile, 12.1 and 18.1 percent to the second and third, and nothing to the highest. These levels are imprecise—a parametric bootstrap puts the lowest quartile’s gain between 8.9 and 45.8 percent—but that the lowest quartile gains most and the highest nothing is not.⁶⁸ The distribution of the gains follows the fertility results: lower costs of use are worth more to the women who already controlled their fertility well, and closing the adherence gap to the women who did not.

Long-acting reversible contraception. At Colorado’s take-up, first births by age 21 fall by 7.1 percent. [Lindo and Packham \(2017\)](#) estimate that the initiative reduced teen birth rates by 6.4 percent over five years in the counties its clinics reached. The model, estimated on a cohort that contracepted before long-acting methods were widely available and given only the program’s take-up, gives an effect of the same order. The two are not the same outcome—the model’s is the share of a cohort with a first birth by 21, the estimate a county-level birth rate among all women under 20—so the comparison checks the order of magnitude, not the digit. As in health, interventions that substitute for the user’s effort close skill gradients that price reductions leave in place ([Goldman and Lakdawalla, 2005](#); [Goldman and Smith, 2002](#)). The gains fall among lower-ability women, where the evaluations of these programs find their effects: the gain is 9.9 percent of lifetime consumption for the lowest quartile and 0.7 for the highest, because removing the adherence margin is worth most to the women for whom adherence was binding. Against closing the adherence gap, a long-acting

⁶⁸Online Appendix [OA.16](#) reports the intervals and the share of draws in which each ordering holds.

method at program scale does less than half as much; offered at the pill’s cost it is taken up by 54 percent of users and lowers first births by 21 by 13 percent, and only a menu in which every method in use is adherence-proof reaches the fall of the targeted experiment, and at 34 percent doubles it. The technology is a substitute for adherence, and at the prices programs have offered it, a partial one. One omission bears on the magnitude: exposure to the risk of conception is held at its observed level, so sexual activity does not respond to a more effective method, and any such response would make the fall smaller.⁶⁹

Schooling and the labor market. The experiments change early fertility far more than they change schooling or careers. Closing the adherence gap lowers first births by age 21 by 16 percent but raises college attendance only from 36.5 to 37.1 percent, and no experiment changes annual earnings by more than \$794 or employment between ages 26 and 45 by more than 1.0 percentage points in any ability quartile. The averted births are concentrated among low-ability women, who study and earn little either way. A teen birth lowers college attendance from 41 to 8 percent in the data, but only 12 percent of women in the lowest quartile attend college at all, and in the model their average earnings at age 30, counting women who are not working, are \$11,375 against \$20,858 in the highest quartile. The gains in Table 7 are therefore the direct value of avoiding an unintended birth, not recovered schooling or earnings. A second reason is a limitation of the model: a woman has at most one child, present for a single period, so an early birth costs one period of reduced labor supply and the experience forgone in it, rather than the extended interruption that [Adda et al. \(2017\)](#) measure (Section 5.3); on this margin the counterfactuals are a lower bound.

⁶⁹Online Appendix [OA.9](#) discusses what adherence and an adherence-proof technology each remove. The women who switch are pill users, whose share the model under-predicts (Section 5.1), so the cost that delivers a given take-up is calibrated on a smaller clinical nest than the data’s.

Table 6. Counterfactual experiments: outcomes

	Baseline	Adherence = Q4	Cost of use		LARC menu		
			No cost of use	Partial cut	Colorado	Pill's cost	All users
<i>Panel A: Aggregate</i>							
First birth by 17	0.141	0.107	0.119	0.124	0.124	0.115	0.073
First birth by 21	0.370	0.310	0.318	0.333	0.344	0.323	0.246
First birth by 25	0.582	0.523	0.512	0.532	0.559	0.534	0.447
Unintended share	0.392	0.334	0.412	0.405	0.365	0.339	0.248
Any method, 18–25	0.741	0.746	0.805	0.786	0.746	0.753	0.759
Pill share among users	0.556	0.532	0.617	0.600	0.322	0.058	0.492
LARC share among users	0.000	0.000	0.000	0.000	0.249	0.540	0.000
College attendance	0.365	0.371	0.381	0.376	0.369	0.375	0.383
Employment, 26–45	0.530	0.530	0.535	0.533	0.531	0.531	0.531
Own earnings at 30	16.108	16.079	16.543	16.409	16.134	16.132	16.092
<i>Panel B: First birth by 21, by AFQT quartile</i>							
Q1	0.536	0.400	0.485	0.500	0.483	0.462	0.340
Q2	0.426	0.372	0.364	0.382	0.404	0.380	0.307
Q3	0.299	0.248	0.246	0.262	0.271	0.248	0.174
Q4	0.222	0.222	0.178	0.190	0.218	0.201	0.161
Cut in cost of use, δ	–	–	100.0%	71.9%	–	–	–

Notes: Experiments as defined in Section 6, run at the estimated parameters with common random numbers. Column headings are abbreviated. ADHERENCE = Q4 gives every quartile the top quartile's adherence; the two cost-of-use columns scale both methods' costs of use κ_n by $1 - \delta$: NO COST OF USE sets $\delta = 1$, and PARTIAL CUT uses the δ that lowers first births by 21 by ten percent; the three LARC columns add an adherence-proof method to the menu at the cost that reproduces Colorado's take-up, at the pill's cost, and to every method in use. Use, pill and long-acting shares are measured over childless woman-periods at ages 18–25; the unintended share is defined as in the estimation moment (Section 3.4). Own earnings at 30 is average annual own wage-and-salary income in thousands of 2016 dollars, taken over all women, including those not working. First births by 17 and the long-acting share, both by ability quartile, are in Online Appendix Table OA.40.

Table 7. Counterfactual experiments: consumption-equivalent variation (percent)

	Adherence = Q4	Cost of use		LARC menu		
		No cost of use	Partial cut	Colorado	Pill's cost	All users
Q1	29.7	46.0	28.1	9.9	18.5	53.8
Q2	12.1	64.8	37.6	3.3	10.3	34.0
Q3	18.1	125.2	63.4	5.7	17.7	67.3
Q4	0.0	140.7	68.9	0.7	8.1	28.7
All	14.7	82.9	46.1	5.0	13.7	44.8

No cost of use, decomposed: the windfall to women already using a method, and the remainder

	Windfall to existing users	Net of windfall
Q1	39.9	6.2
Q2	53.8	11.0
Q3	95.5	29.6
Q4	104.5	36.1

Notes: Permanent proportional change in consumption that makes a woman at age 14 indifferent between the baseline and the experiment, from the value function by ability quartile; the last row weights quartiles by population. Columns are the experiments of Table 6. Lower panel, NO COST OF USE only: *windfall to existing users* is the expected discounted saving on the methods a woman chose at baseline when their costs of use are removed, which she collects without changing any choice; *net of windfall* is the total less the windfall, both in consumption units, so the two add up to the total: the value of the fertility control the lower cost induces and everything downstream of it. Online Appendix OA.15 reports the full decomposition, including the partial cut.

7 Conclusion

This paper asks whether the channels standard life-cycle models emphasize, schooling and opportunity costs, can explain why women with higher cognitive ability delay first births. In the NLSY79 the ability gradient in early fertility is steep and closes with age, and the additional first births of low-ability women are births they did not want at that time. The gradient opens while girls are still in high school and survives holding their schooling to date and their 1979 expectations fixed; in the pooled population it persists within contraceptive method, and method selection runs *counter* to it. What remains is fertility control: ability predicts whether a method is used and, among users, how well it works, and predicts nothing about conception among women using none.

A life-cycle model in which cognitive ability shifts adherence reproduces the gradient and its convergence with age while matching education, marriage, labor supply and contraception. Within this model, opportunity costs and preferences alone do not rationalize the data: common adherence is rejected. Without an ability difference in adherence the model spreads teen births up the ability distribution, cannot produce the ability gradient in failure among users of the same method, recovers only 79 percent of the gradient in unintended first births by 21, and distorts labor supply to keep births where the timing moments want them. The estimated adherence gradient is large at ages 14–25 and closed by 26, where the ability gradient in first-birth hazards reverses.

Equalizing adherence lowers first births by age 21 by 16 percent and is worth 30 percent of lifetime consumption to the lowest ability quartile against nothing to the highest. Removing the cost of using contraception, an upper bound on what any subsidy can do, lowers first births by 21 by 14 percent, about as much as closing the adherence gap, but its gains accrue to the top of the ability distribution. A long-acting method added to the menu at Colorado’s take-up lowers first births by 21 by 7 percent, close to what that program’s evaluation found, with the largest gains in the lowest quartile and the smallest in the highest; only a menu in which every user’s method is adherence-proof does as much as equalizing adherence, or more. For disadvantaged women in this cohort, and within this model, the margin with more room is the capacity to control fertility timing rather than the incentive to do so.

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